



STP Principles and Configuration



Foreword

- On an Ethernet switching network, redundant links are used to implement link backup and enhance network reliability. However, the use of redundant links may produce loops, leading to broadcast storms and an unstable MAC address table. As a result, communication on the network may deteriorate or even be interrupted. To prevent loops, IEEE introduced the Spanning Tree Protocol (STP).
- Devices running STP exchange STP Bridge Protocol Data Units (BPDUs) to discover loops on the network and block appropriate ports. This enables a ring topology to be trimmed into a loop-free tree topology, preventing infinite looping of packets and ensuring packet processing capabilities of devices.
- IEEE introduced the Rapid Spanning Tree Protocol (RSTP) to improve the network convergence speed.



Objectives

- Upon completion of this course, you will be able to:
 - Describe the causes and problems of Layer 2 loops on a campus switching network.
 - Describe basic concepts and working mechanism of STP.
 - Distinguish STP from RSTP and describe the improvement of RSTP on STP.
 - Complete basic STP configurations.
 - Understand other methods to eliminate Layer 2 loops on the switching network except STP.



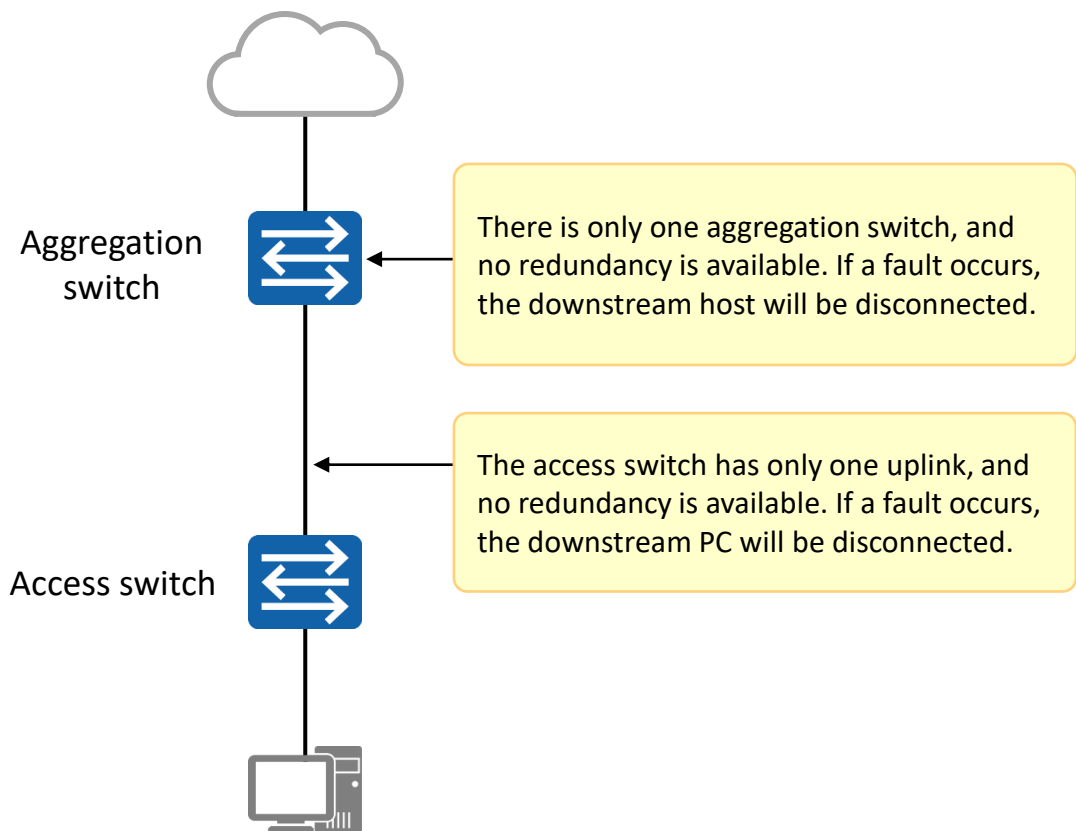
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- 1. STP Overview**
2. Basic Concepts and Working Mechanism of STP
3. Basic STP Configurations
4. Improvements Made in RSTP
5. STP Advancement

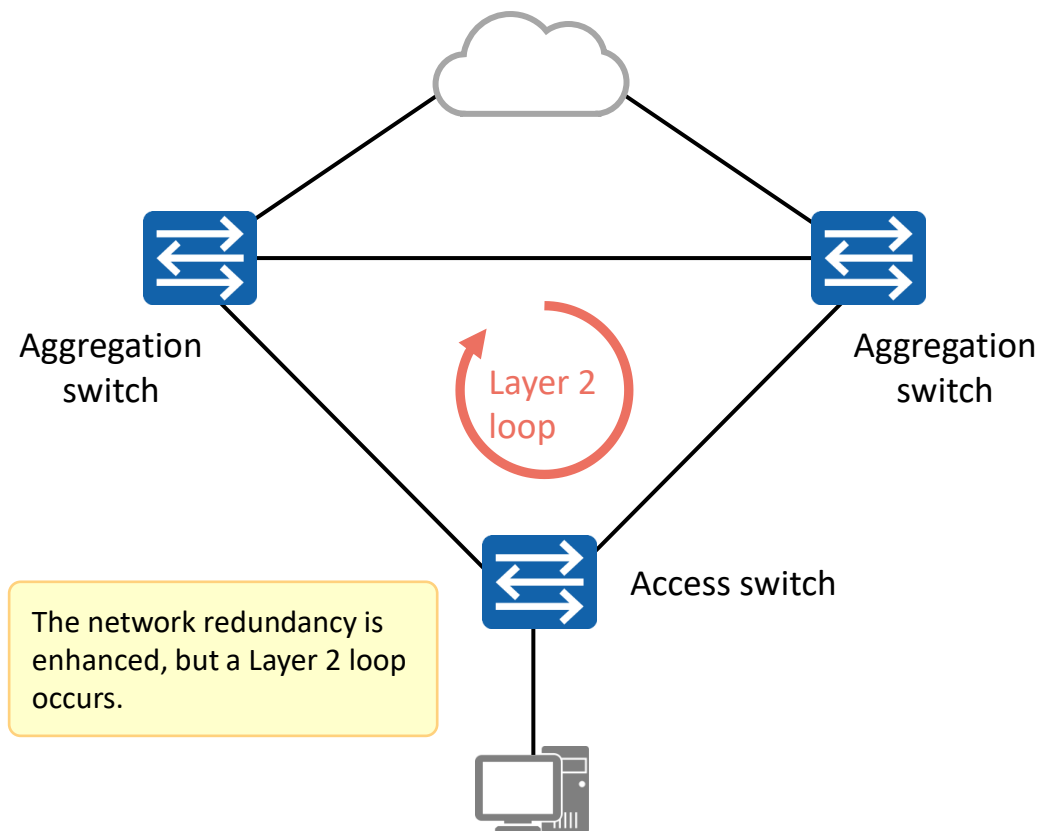


Technical Background: Redundancy and Loops on a Layer 2 Switching Network

A network without redundancy design



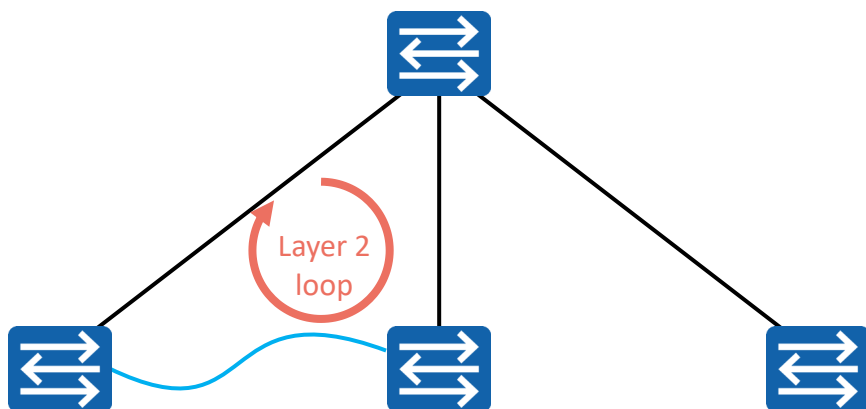
Layer 2 loops introduced along with redundancy





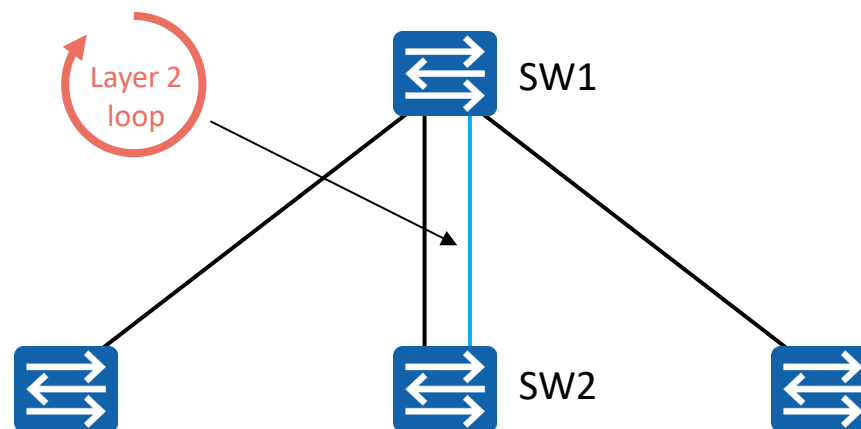
Technical Background: Layer 2 Loops Caused by Human Errors

Case 1



Incorrect operations: For example, connections of cables between devices are incorrect.

Case 2

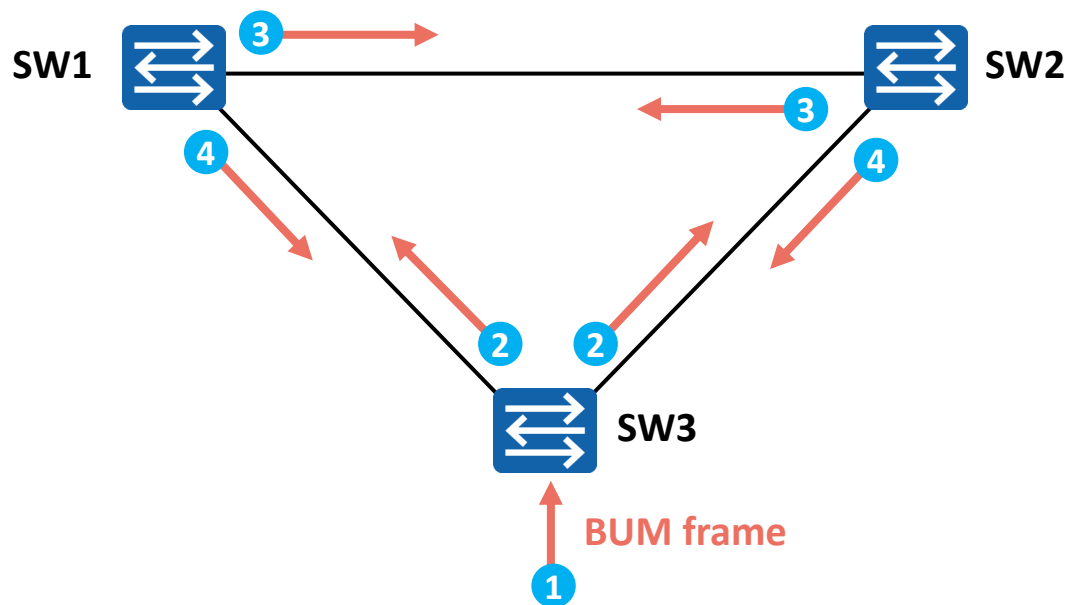


Incorrect manual configurations: For example, the network administrator does not bind the link between SW1 and SW2 to a logical link (aggregation link), causing Layer 2 loops.



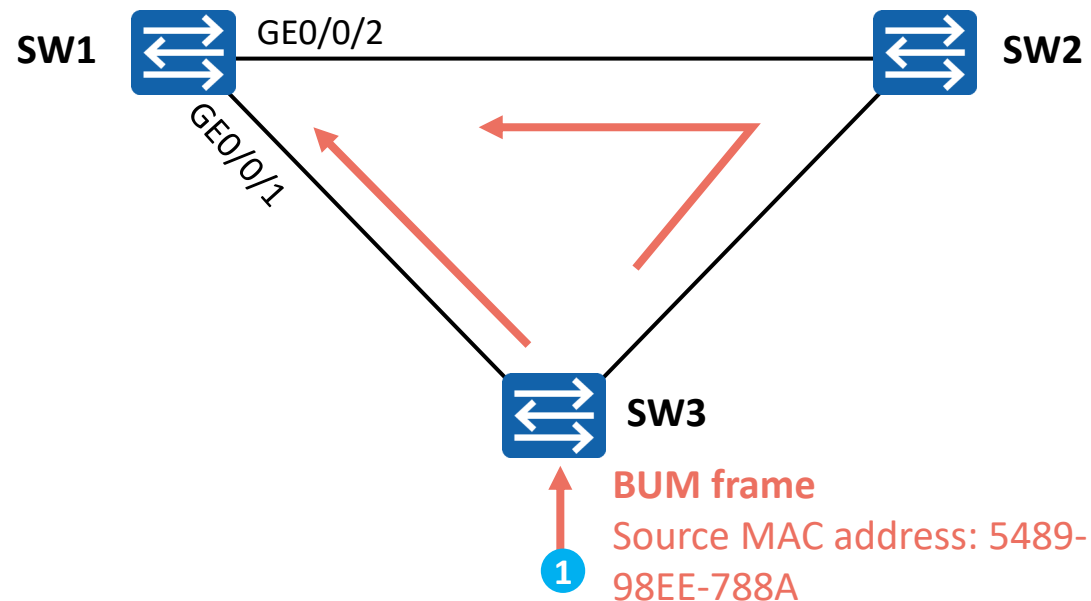
Issues Caused by Layer 2 Loops

Typical Issue 1: Broadcast Storm



When SW3 receives the BUM frames, it floods the frames. After SW1 and SW2 receive the BUM frames, they flood the frames again. As a result, network resources are exhausted and the network is unavailable.

Typical Issue 2: MAC Address Flapping

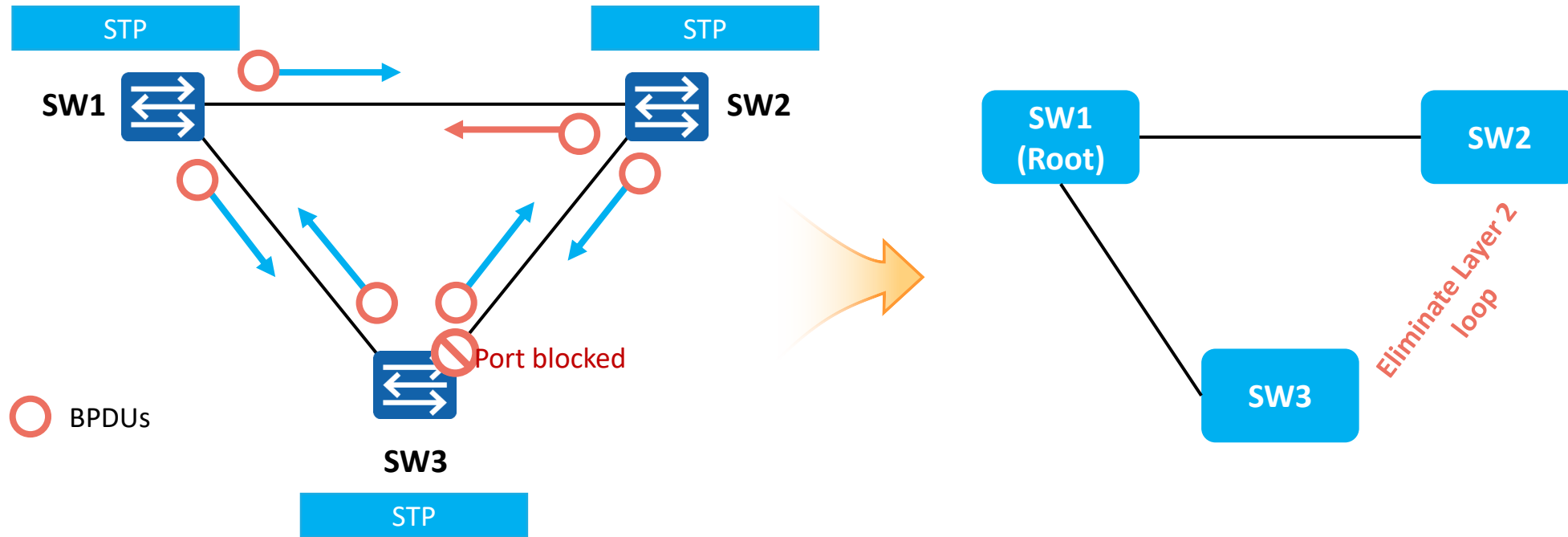


SW1 is used as an example. The MAC address of 5489-98EE-788A is frequently switched between GE0/0/1 and GE0/0/2, causing MAC address flapping.

BUM frames: broadcast, unknown unicast, and multicast frames



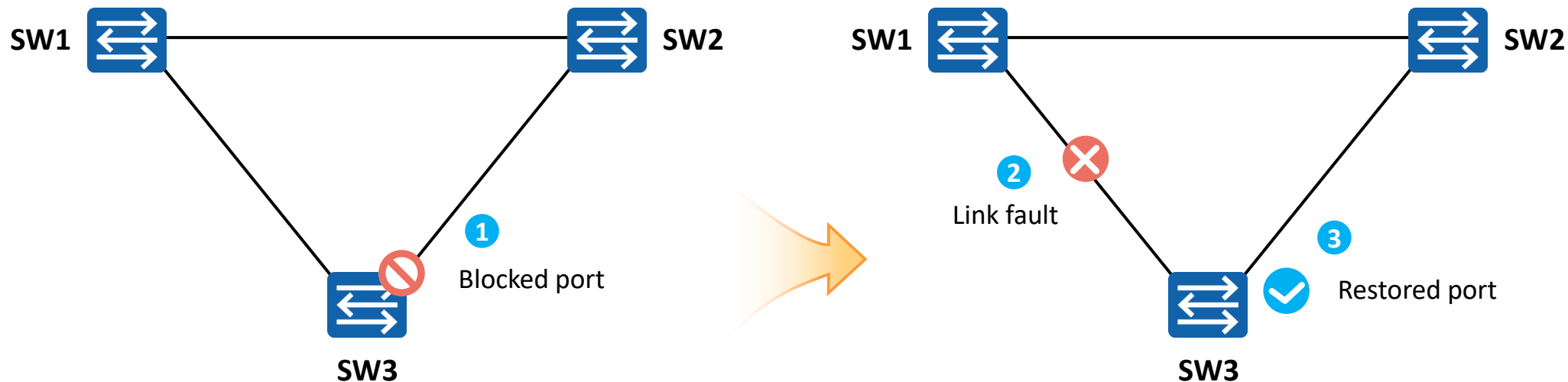
Introduction to STP



When STP is deployed on a network, switches exchange STP BPDUs and calculate a loop-free topology. Finally, one or more ports on the network are blocked to eliminate loops.



STP Can Dynamically Respond to Network Topology Changes and Adjust Blocked Ports



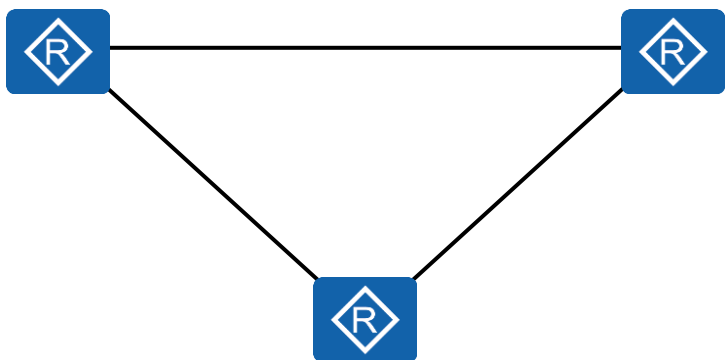
STP running on a switch continuously monitors the network topology. When the network topology changes, STP can detect the changes and automatically adjust the network topology.

Therefore, STP can solve the Layer 2 loop problem and provide a solution for network redundancy.



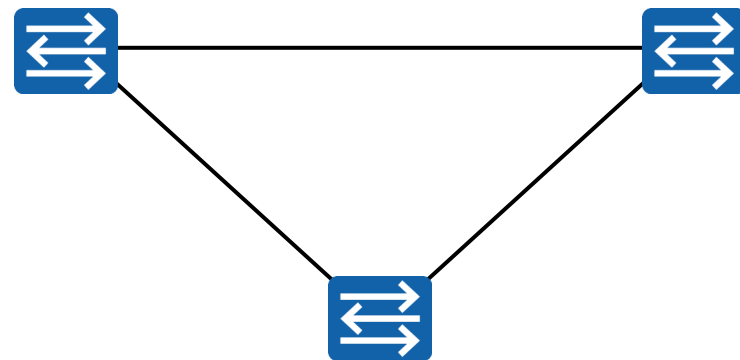
Q&A: Layer 2 and Layer 3 loops

Layer 3 loop



- Common root cause: routing loop
- Dynamic routing protocols have certain loop prevention capabilities.
- The TTL field in the IP packet header can be used to prevent infinite packet forwarding.

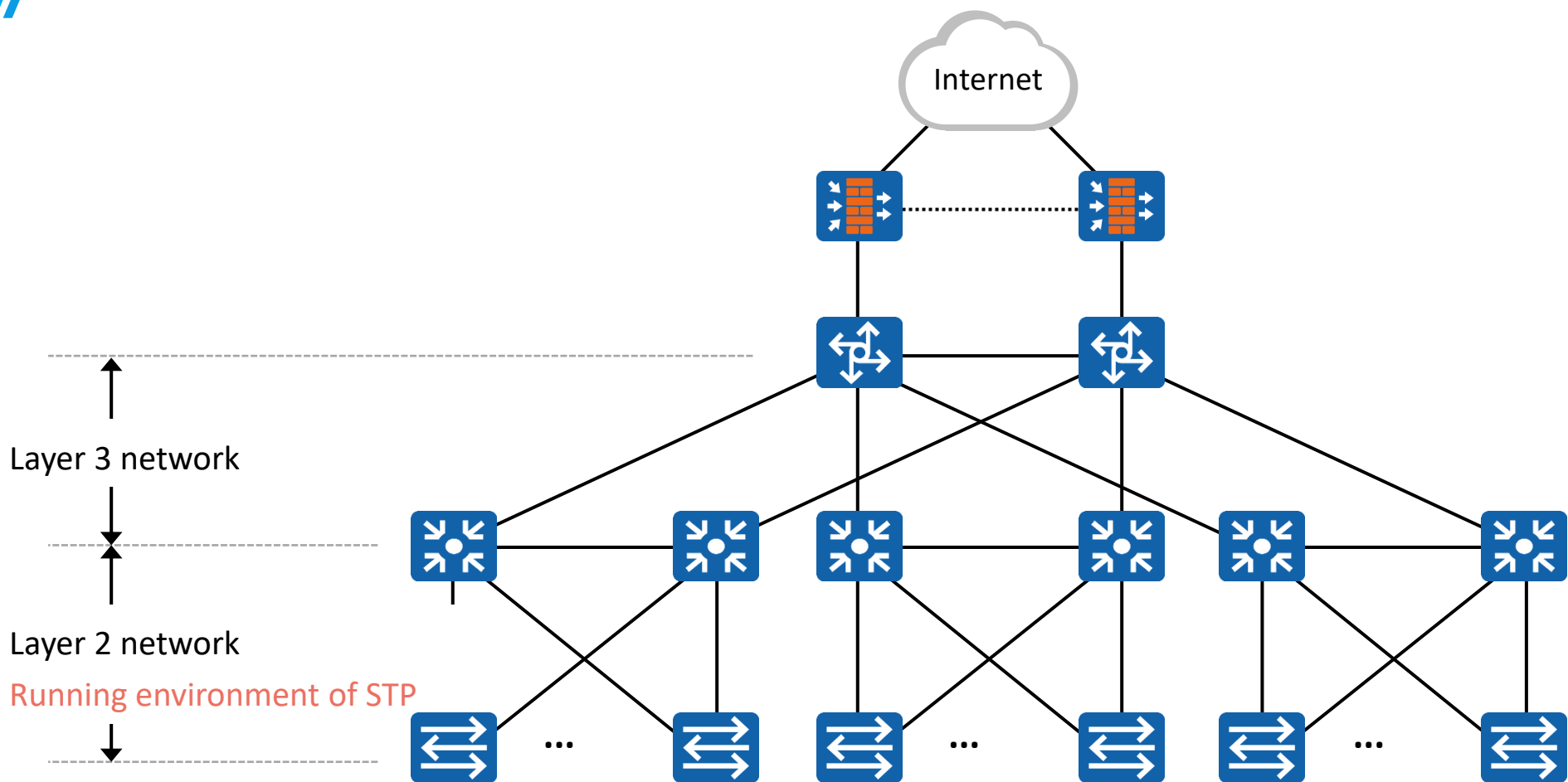
Layer 2 loop



- Common root cause: Layer 2 redundancy is deployed on the network, or cables are incorrectly connected.
- Specific protocols or mechanisms are required to implement Layer 2 loop prevention.
- The Layer 2 frame header does not contain any information to prevent data frames from being forwarded infinitely.



Application of STP on a Campus Network





STP Overview

- STP is used on a LAN to prevent loops.
- Devices running STP exchange information with one another to discover loops on the network, and block certain ports to eliminate loops.
- After running on a network, STP continuously monitors the network status. When the network topology changes, STP can detect the change and automatically respond to the change. In this way, the network status can adapt to the new topology, ensuring network reliability.
- With the growth in scale of LANs, STP has become an important protocol for a LAN.



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- 2. Basic Concepts and Working Mechanism of STP**
3. Basic STP Configurations
4. Improvements Made in RSTP
5. STP Advancement



STP Basic Concepts: BID

BID

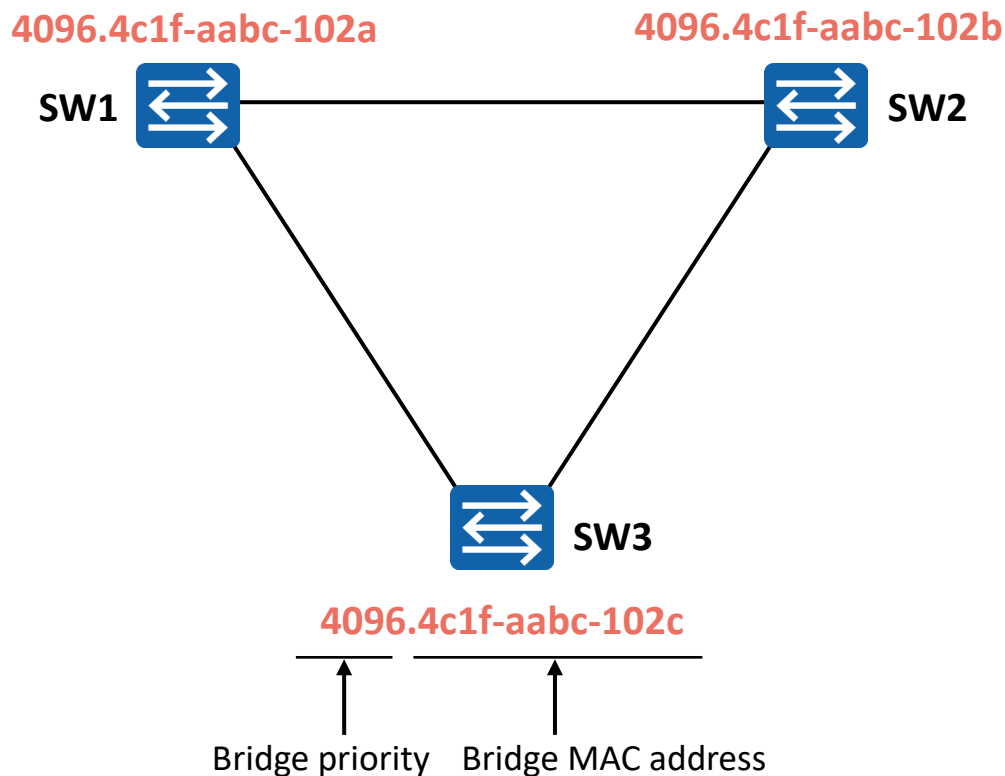
Root
Bridge

Cost

RPC

PID

BPDU



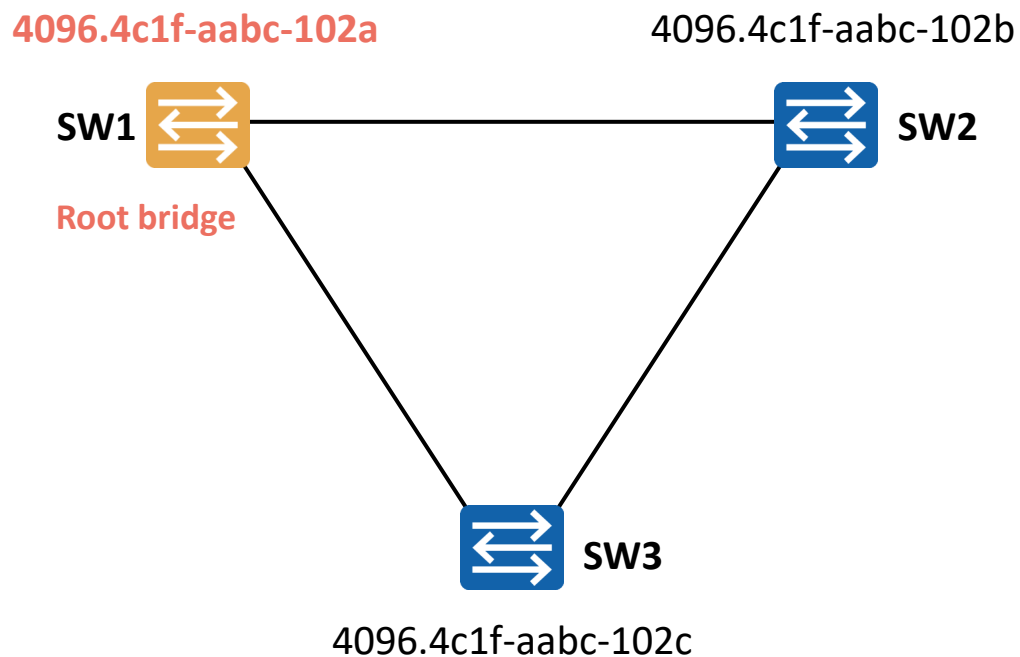
Bridge ID (BID)

- As defined in IEEE 802.1D, a BID consists of a 16-bit bridge priority and a bridge MAC address.
- Each switch running STP has a unique BID.
- The bridge priority occupies the leftmost 16 bits and the MAC address occupies the rightmost 48 bits.
- On an STP network, the device with the smallest BID acts as the root bridge.

Note: A bridge is a switch.



STP Basic Concepts: Root Bridge



Root Bridge

- One of the main functions of STP is to calculate a loop-free STP tree on the entire switching network.
- The root bridge is the root of an STP network.
- After STP starts to work, it elects a root bridge on the switching network. The root bridge is the key for topology calculation of the spanning tree and is the root of the loop-free topology calculated by STP.
- On an STP network, the device with the smallest BID acts as the root bridge.

During BID comparison, devices first compare bridge priorities. A smaller priority value indicates a higher priority of a device. The switch with the smallest priority value becomes the root bridge. If priority values are the same, the switch with the smallest MAC address becomes the root bridge.



STP Basic Concepts: Cost

BID

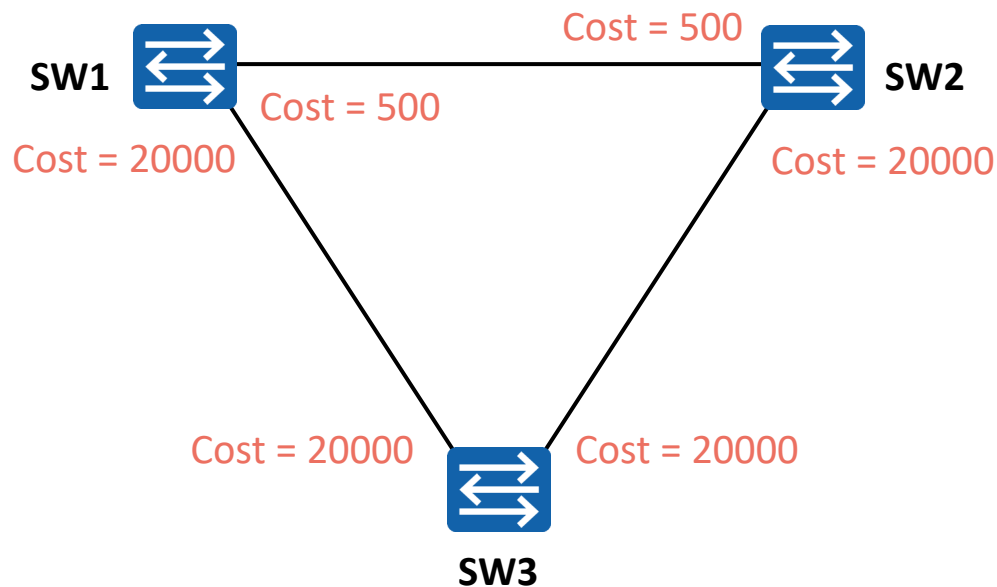
Root
Bridge

Cost

RPC

PID

BPDU



Cost

- Each STP-enabled port maintains a cost. The cost of a port is used to calculate the root path cost (RPC), that is, the cost of the path to the root.
- The default cost of a port is related to the rate, working mode, and STP cost calculation method used by a switch.
- A higher port bandwidth indicates a smaller cost.
- You can also run commands to adjust the cost of a port as required.



STP Basic Concepts: Cost Calculation Methods

Port Rate	Port Mode	Recommended STP Cost		
		IEEE 802.1d-1998	IEEE 802.1t	Huawei Legacy Standard
100 Mbit/s	Half-duplex	19	200,000	200
	Full-duplex	18	199,999	199
	Aggregated link: two ports	15	100,000	180
1000 Mbit/s	Full-duplex	4	20,000	20
	Aggregated link: two ports	3	10,000	18
10 Gbit/s	Full-duplex	2	2000	2
	Aggregated link: two ports	1	1000	1
40 Gbit/s	Full-duplex	1	500	1
	Aggregated link: two ports	1	250	1
100 Gbit/s	Full-duplex	1	200	1
	Aggregated link: two ports	1	100	1
...				

The cost has a default value and is associated with the port rate. When the device uses different algorithms, the same port rate corresponds to different cost values.



STP Basic Concepts: RPC

BID

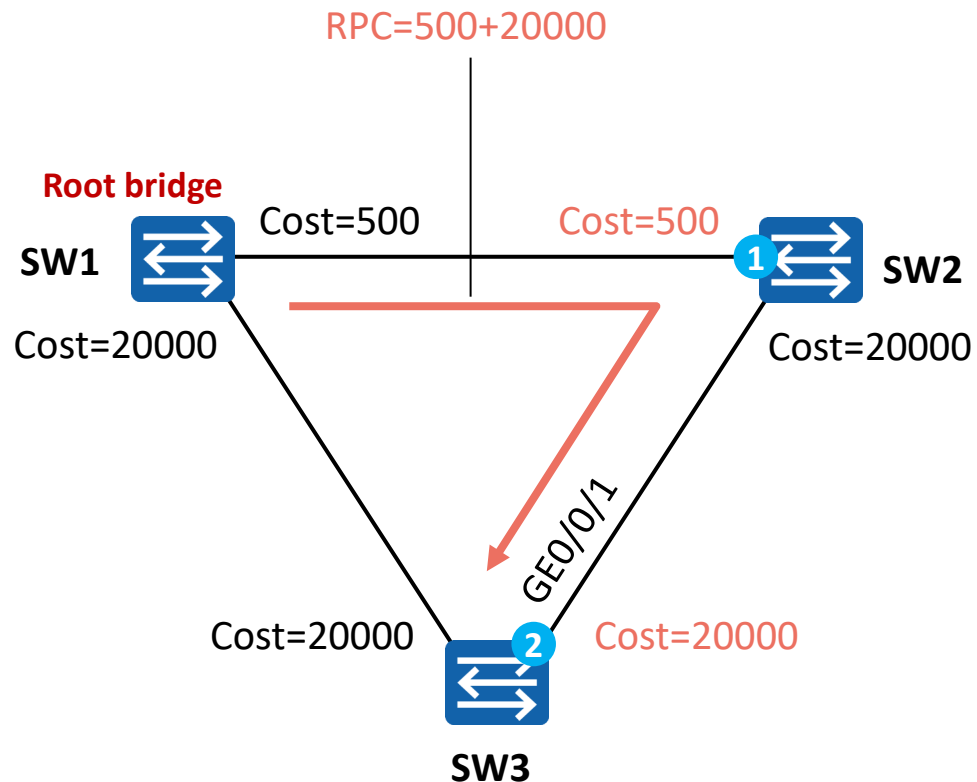
Root
Bridge

Cost

RPC

PID

BPDU



RPC

- The cost from a switch port to the root bridge, that is, RPC, is important during STP topology calculation.
- The RPC from a port to the root bridge is the sum of costs of all inbound ports along the path from the root bridge to the device.
- In this example, the RPC for SW3 to reach the root bridge through GE0/0/1 is equal to the cost of port 1 plus the cost of port 2.



STP Basic Concepts: PID

BID

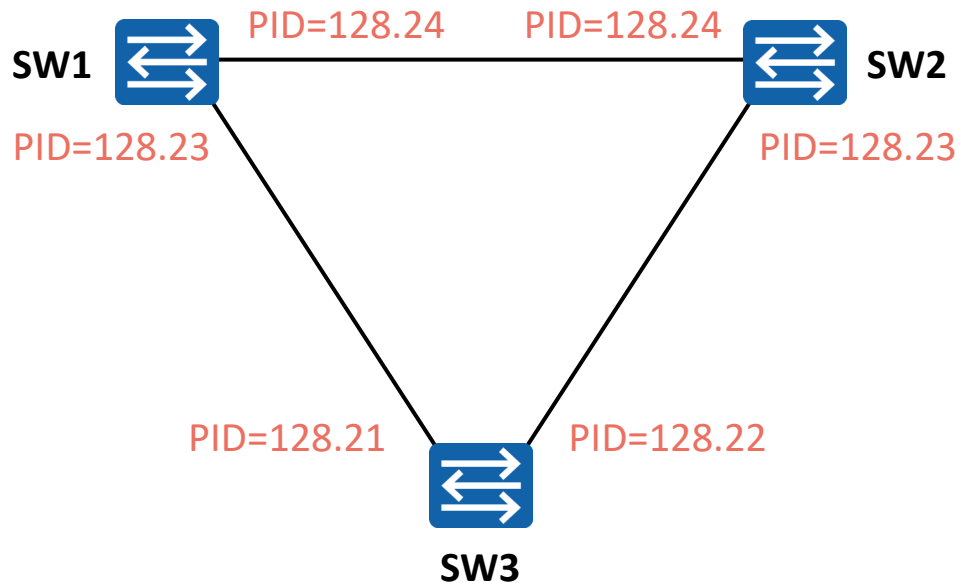
Root
Bridge

Cost

RPC

PID

BPDU



Port ID (PID)

- An STP-enabled switch uses PIDs to identify ports. A PID is used to elect a designated port in a specific scenario.
- A PID consists of the leftmost four bits (port priority) and the rightmost 12 bits (port number).
- An STP-enabled port maintains a default port priority, which is 128 on Huawei switches. You can run a command to change the priority as required.



STP Basic Concepts: BPDU

BID

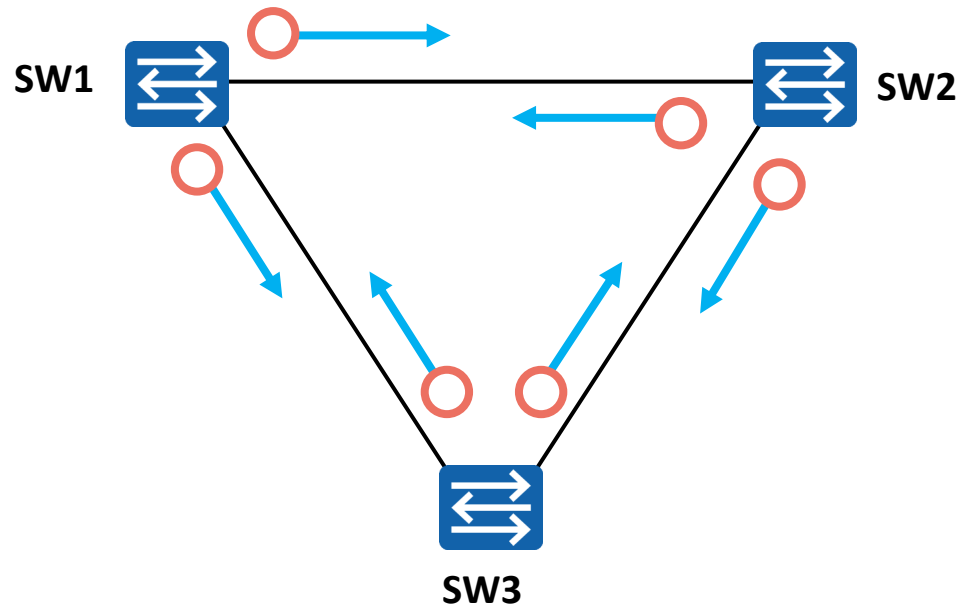
Root
Bridge

Cost

RPC

PID

BPDU



○ Configuration BPDU

Bridge Protocol Data Unit (BPDU)

- BPDU is the basis for STP to work normally.
- STP-enabled switches exchange BPDUs that carry important information.
- There are two types of BPDUs:
 - Configuration BPDU
 - Topology Change Notification (TCN) BPDU
- Configuration BPDUs are the key to STP topology calculation. TCN BPDUs are triggered only when the network topology changes.



Format of Configuration BPDUs

PID	PVI	BPDU Type	Flags	Root ID	RPC	Bridge ID	Port ID	Message Age	Max Age	Hello Time	Forward Delay
-----	-----	-----------	-------	---------	-----	-----------	---------	-------------	---------	------------	---------------

Byte	Field	Description
2	PID	For STP, the value of this field is always 0.
1	PVI	For STP, the value of this field is always 0.
1	BPDU Type	Type of BPDUs. The value 0x00 indicates a configuration BPDU and the value 0x80 indicates a TCN BPDU.
1	Flags	STP uses only the leftmost two bits and the rightmost two bits: Topology Change Acknowledgment (TCA) and Topology Change (TC).
8	Root D	BID of the root bridge.
4	RPC	STP cost of the path from the current port to the root bridge.
8	Bridge ID	BID of the sender.
2	Port ID	ID of the port that sends this BPDU, which consists of the port priority and port number.
2	Message Age	Number of seconds after a BPDU is sent from the root bridge. The value increases by 1 each time the BPDU passes through a network bridge. It refers to the number of hops to the root bridge.
2	Max Age	If the bridge does not receive any BPDU for a period of time and the lifetime of the network bridge reaches the maximum, the network bridge considers that the link connected to the port is faulty. The default value is 20s.
2	Hello Time	Interval at which the root bridge sends configuration BPDUs. The default value is 2s.
2	Forward Delay	Time that is spent in Listening or Learning state. The default value is 15s.



BPDU Comparison Rules

Field

Protocol Identifier

Protocol Version Identifier

BPDU Type

Flags

Root Identifier

Root Path Cost

Bridge Identifier

Port Identifier

Message Age

Max Age

Hello Time

Forward Delay

The core of STP is to calculate a loop-free topology on a switching network. During topology calculation, the comparison of configuration BPDUs is important. The Root Identifier, Root Path Cost, Bridge Identifier, and Port Identifier fields are the main fields of a configuration BPDU. STP-enabled switches compare the four fields.

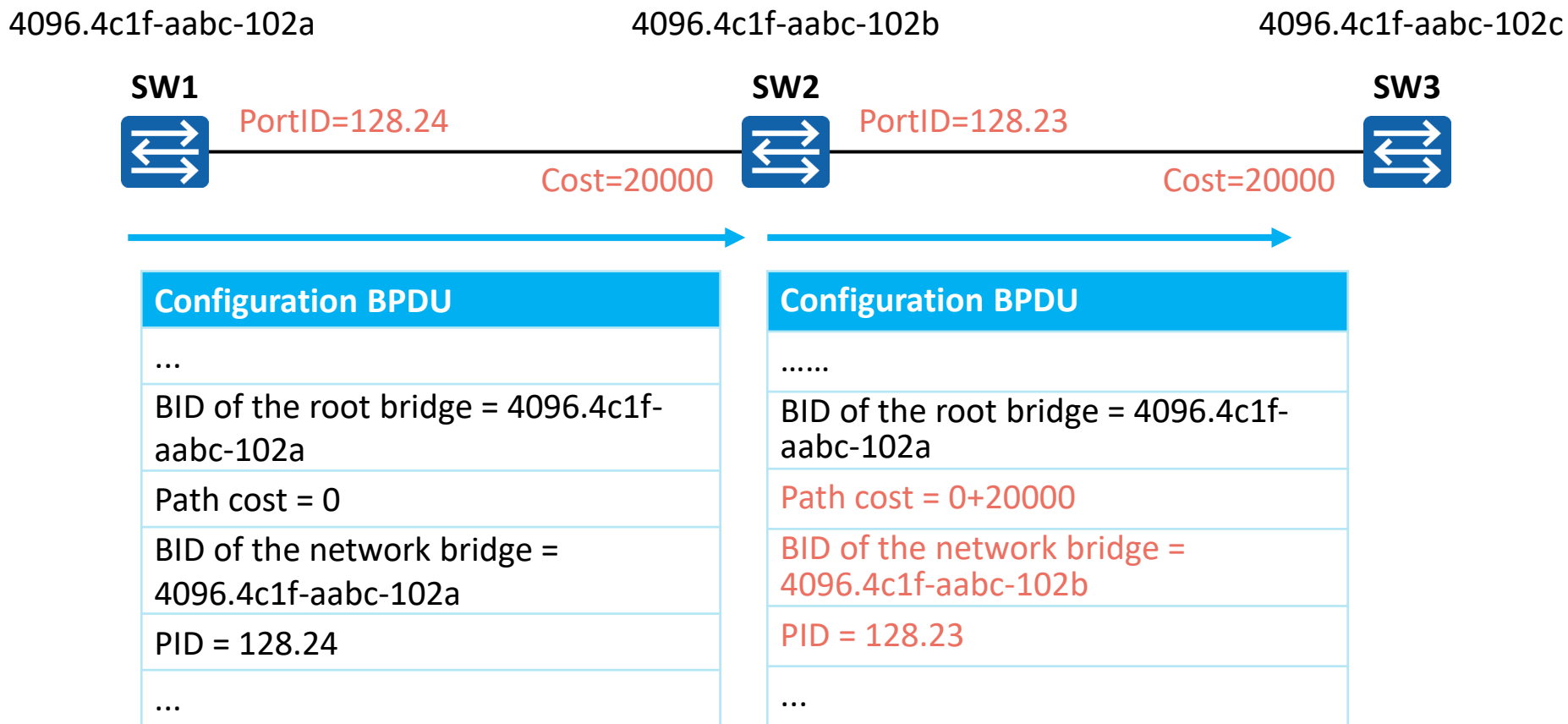
STP selects the optimal configuration BPDU in the following sequence:

1. Smallest BID of the root bridge
2. Smallest RPC
3. Smallest BID of the network bridge
4. Smallest PID

Among the four rules (each rule corresponds to a field in a configuration BPDU), the first rule is used to elect the root bridge on the network, and the following rules are used to elect the root port and designated port.



Configuration BPDU Forwarding Process





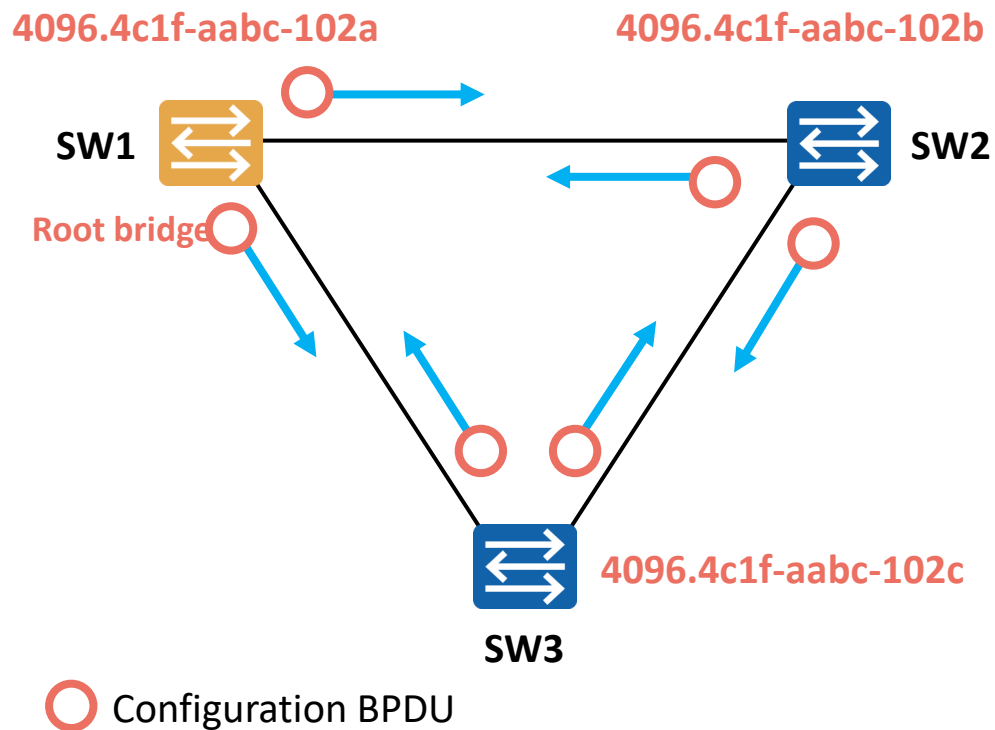
STP Calculation (1)

Select Root
Bridge

Select Root Port

Select Designated
Port

Block Non-designated
Port



Select a Root Bridge on the Switching Network

- After STP starts to work on a switching network, each switch sends configuration BPDUs to the network. The configuration BPDU contains the BID of a switch.
- The switch with the smallest bridge ID becomes the root bridge.
- Only one root bridge exists on a contiguous STP switching network.
- The role of the root bridge can be preempted.
- To ensure the stability of the switching network, you are advised to plan the STP network in advance and set the bridge priority of the switch that is planned as the root bridge to the minimum value 0.



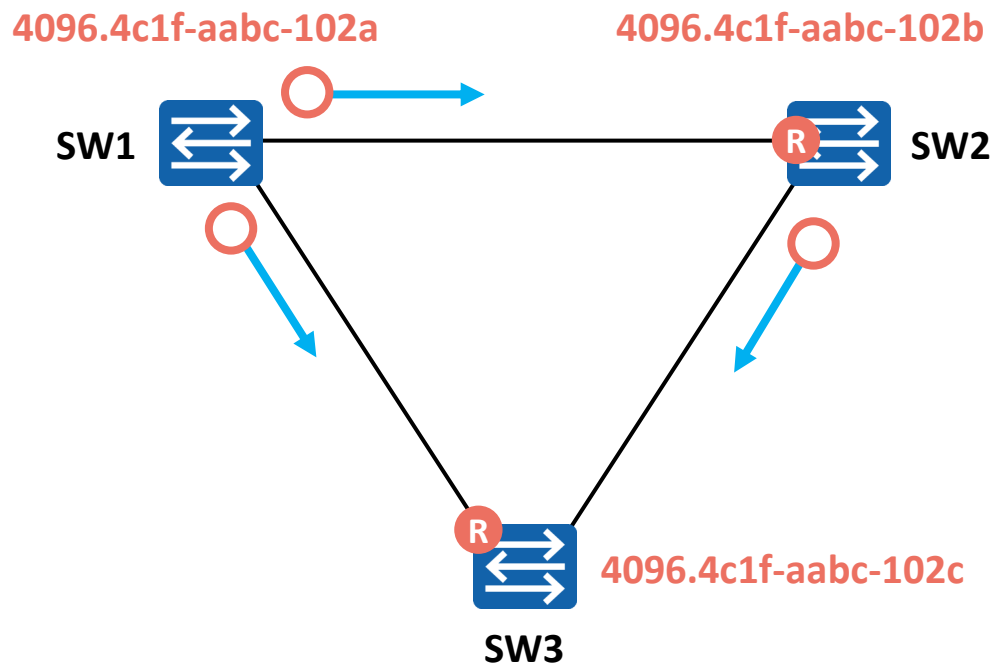
STP Calculation (2)

Select Root Bridge

Select Root Port

Select Designated Port

Block Non-designated Port



○ Configuration BPDU R Root port

Select a Root Port on Each Non-root Bridge

- Each non-root bridge selects a root port from its ports.
- A non-root bridge has only one root port.
- When a non-root-bridge switch has multiple ports connected to the network, the root port receives the optimal configuration BPDU.
- The root port is located on each non-root bridge and has the shortest distance away from the root bridge.



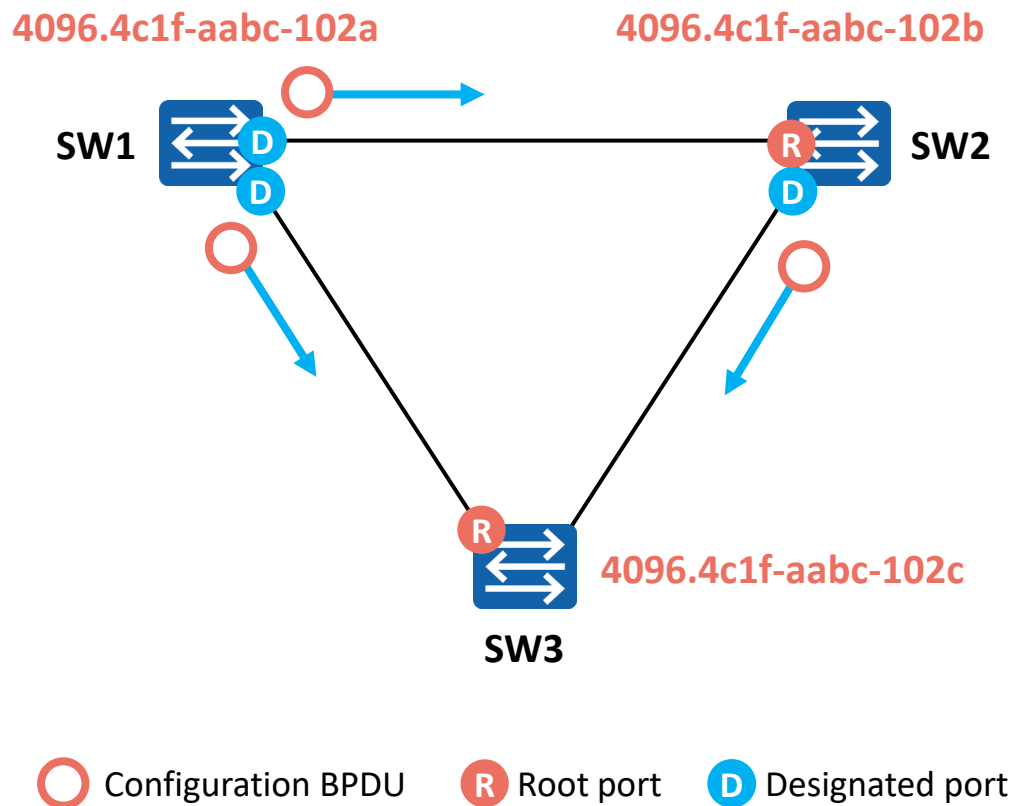
STP Calculation (3)

Select Root Bridge

Select Root Port

Select Designated Port

Block Non-designated Port



A designated port is elected on each link.

- After the root port is elected, the non-root bridge uses the optimal BPDU received on the port to calculate the configuration BPDU and compares the calculated configuration BPDU with the configuration BPDUs received by all ports except the root port.
 - If the former is better, the port is a designated port.
 - If the latter is better, the port is not a designated port.
- In most cases, all ports on the root bridge are designated ports.



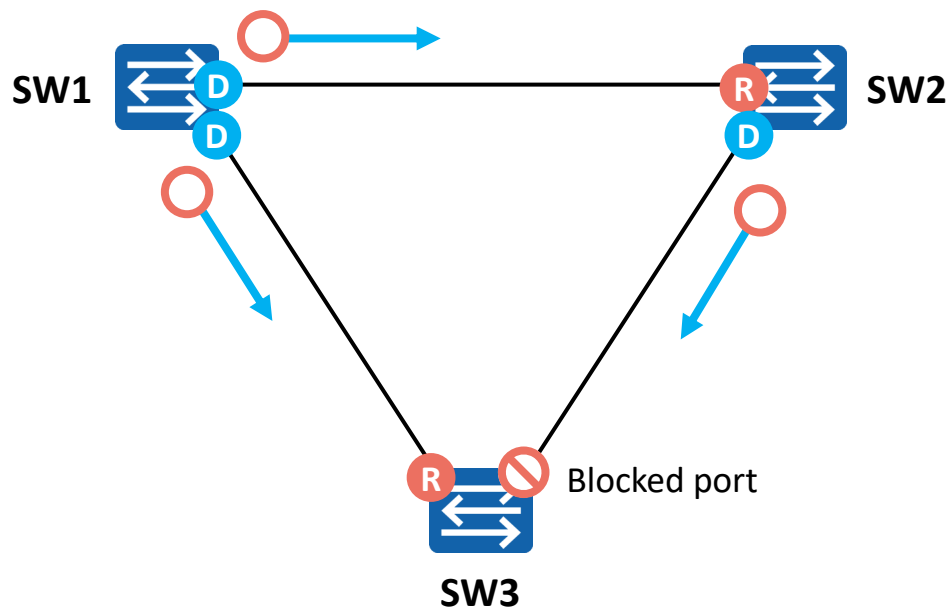
STP Calculation (4)

Select Root Bridge

Select Root Port

Select Designated Port

Block Non-designated Port



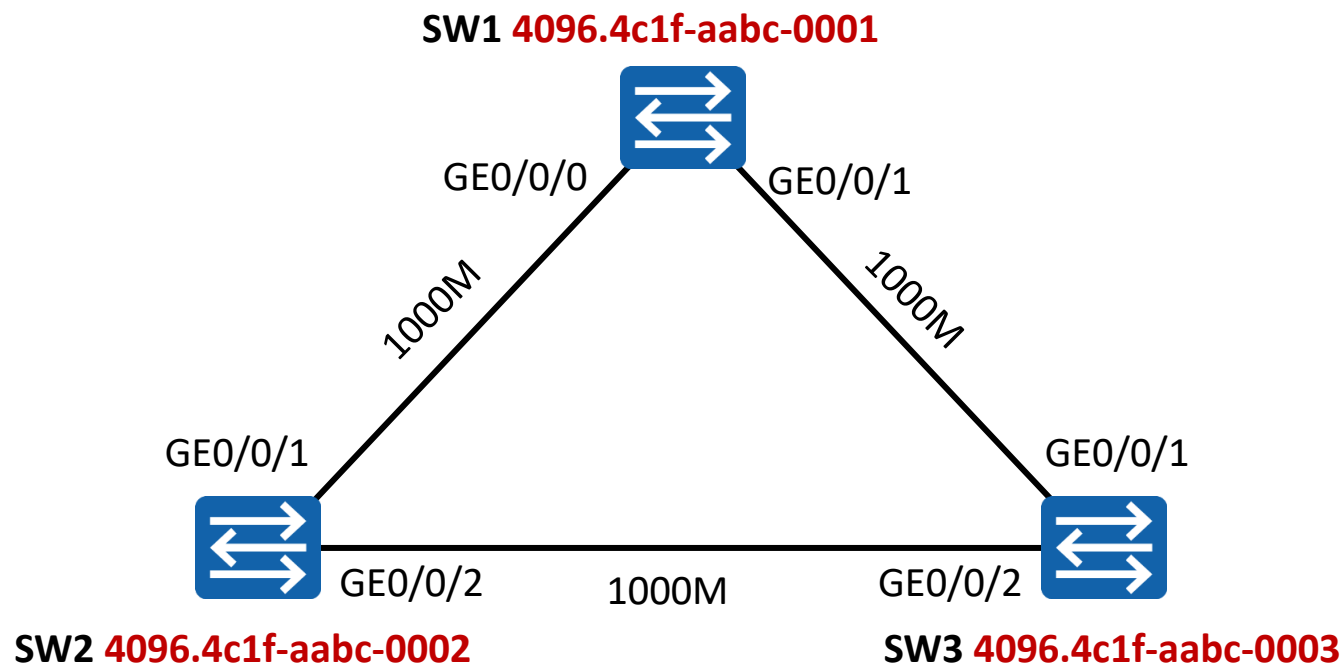
○ Configuration BDU ● Root port ● Designated port

Block Non-designated Port

- On a switch, a port that is neither a root port nor a designated port is called a non-designated port.
- The last step of STP operations is to block the non-designated port on the network. After this step is complete, the Layer 2 loop on the network is eliminated.

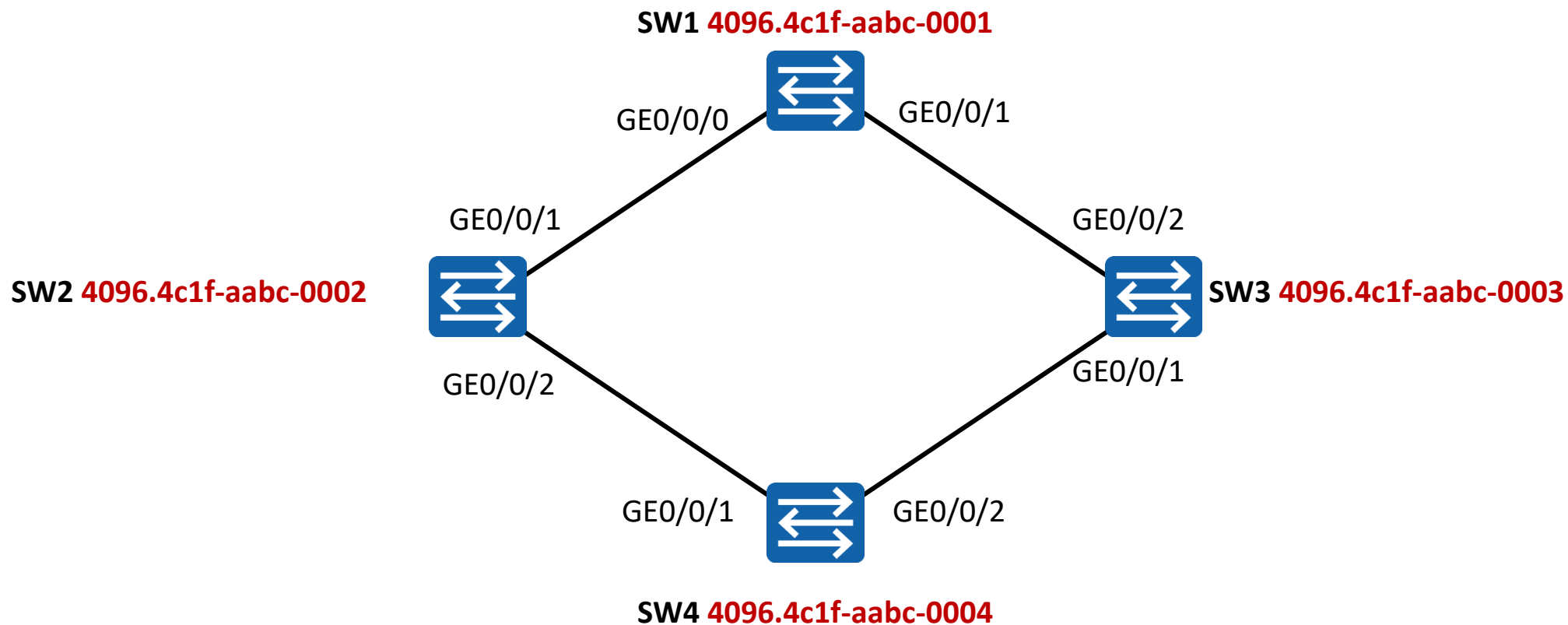


Quiz 1: Identify the Root Bridge and Port Roles



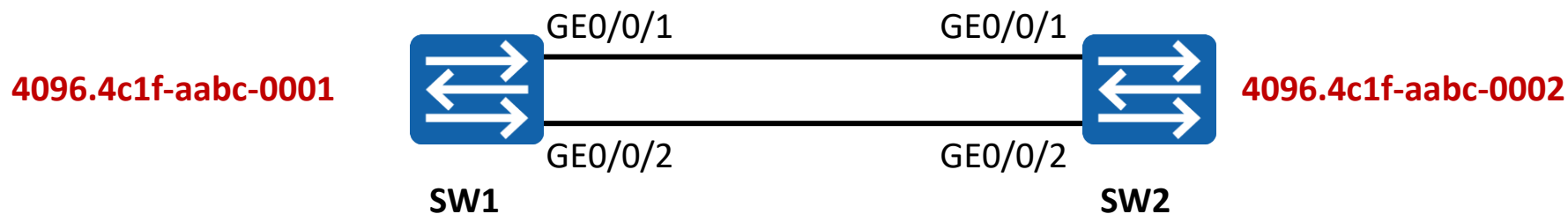


Quiz 2: Identify the Root Bridge and Port Roles in the Following Topology





Quiz 3: Identify the Root Bridge and Port Roles in the Following Topology



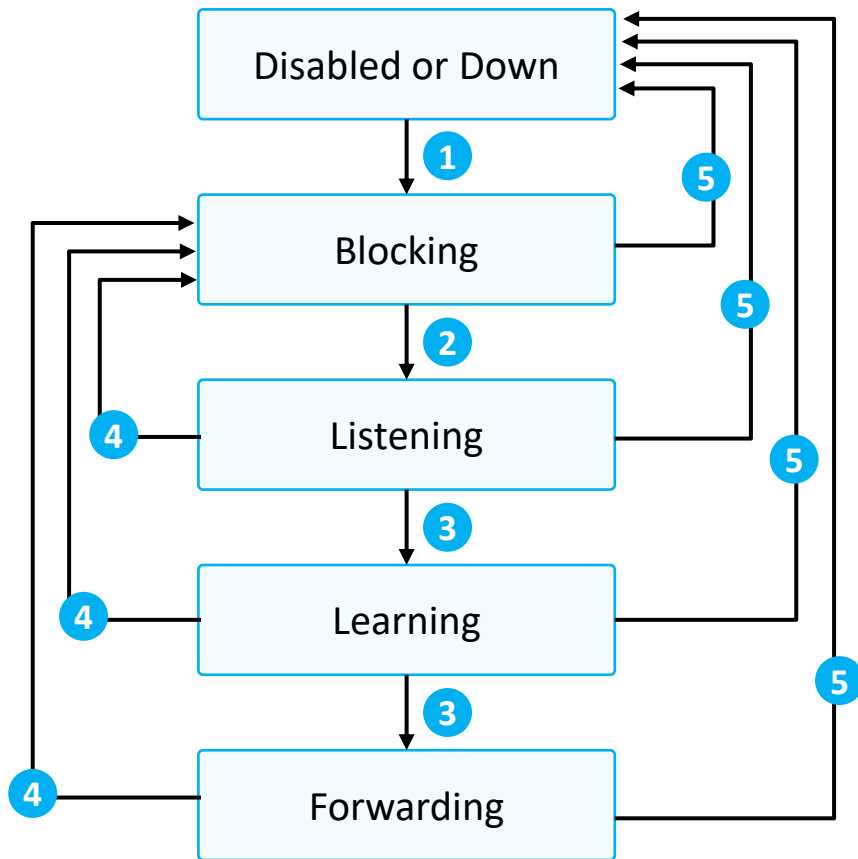


STP Port States

Port State	Description
Disabled	The port cannot send or receive BPDUs or service data frames. That is, the port is Down.
Blocking	The port is blocked by STP. A blocked port cannot send BPDUs but listens to BPDUs. In addition, the blocked port cannot send or receive service data frames or learn MAC addresses.
Listening	STP considers the port in Listening state as the root port or designated port, but the port is still in the STP calculation process. In this case, the port can send and receive BPDUs but cannot send or receive service data frames or learn MAC addresses.
Learning	A port in Learning state listens to service data frames but cannot forward them. After receiving service data frames, the port learns MAC addresses.
Forwarding	A port in Forwarding state can send and receive service data frames and process BPDUs. Only the root port or designated port can enter the Forwarding state.



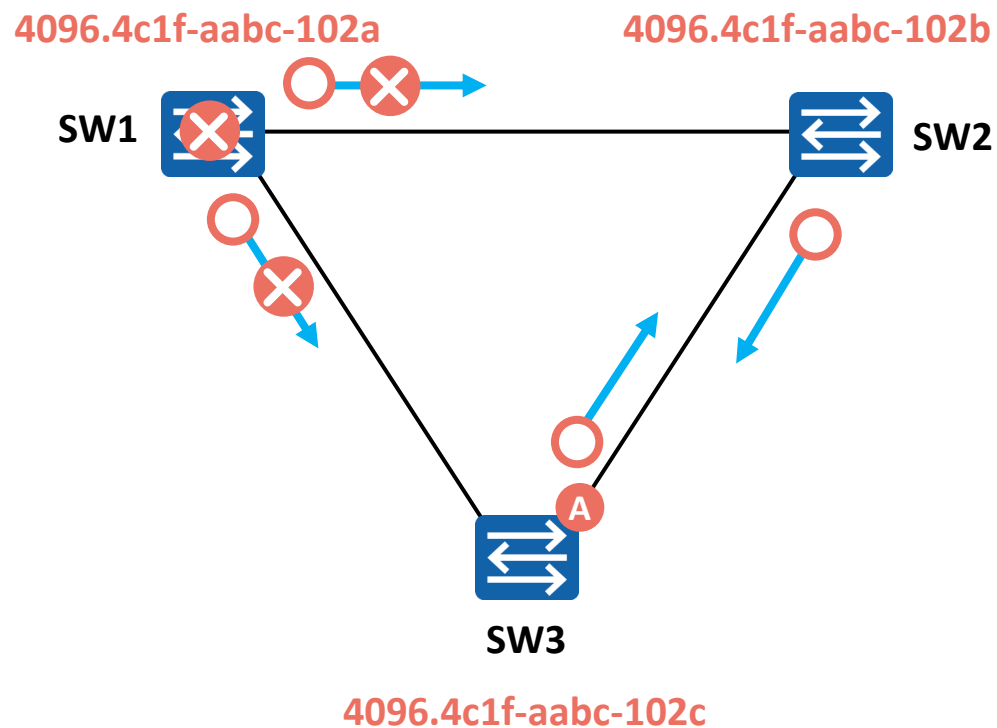
STP Port State Transition



- 1 When a port is initialized or activated, it automatically enters the blocking state.
- 2 The port is elected as the root port or designated port and automatically enters the Listening state.
- 3 The Forward Delay timer expires and the port is still the root port or designated port.
- 4 The port is no longer the root port or designated port.
- 5 The port is disabled or the link fails.



Topology Change: Root Bridge Fault

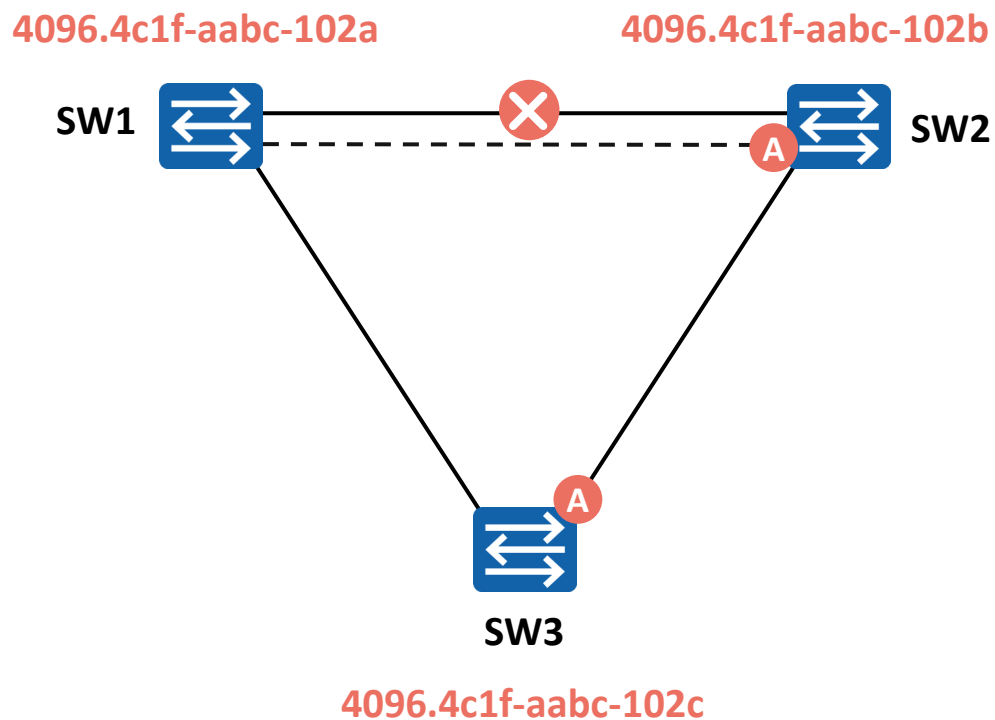


Root Bridge Fault Rectification Process

1. SW1 (root bridge) is faulty and stops sending BPDUs.
2. SW2 waits for the Max Age timer (20s) to expire. In this case, the record about the received BPDUs becomes invalid, and SW2 cannot receive new BPDUs from the root bridge. SW2 learns that the upstream device is faulty.
3. Non-root bridges send configuration BPDUs to each other to elect a new root bridge.
4. After re-election, port A of SW3 transitions to the Forwarding state after two intervals of the Forward Delay timer (the default interval is 15s).
 - A non-root bridge starts root bridge re-election after BPDUs age.
 - Due to the root bridge failure, it takes about 50s to recover from a root bridge failure.



Topology Change: Direct Link Fault



Direct Link Fault Rectification Process

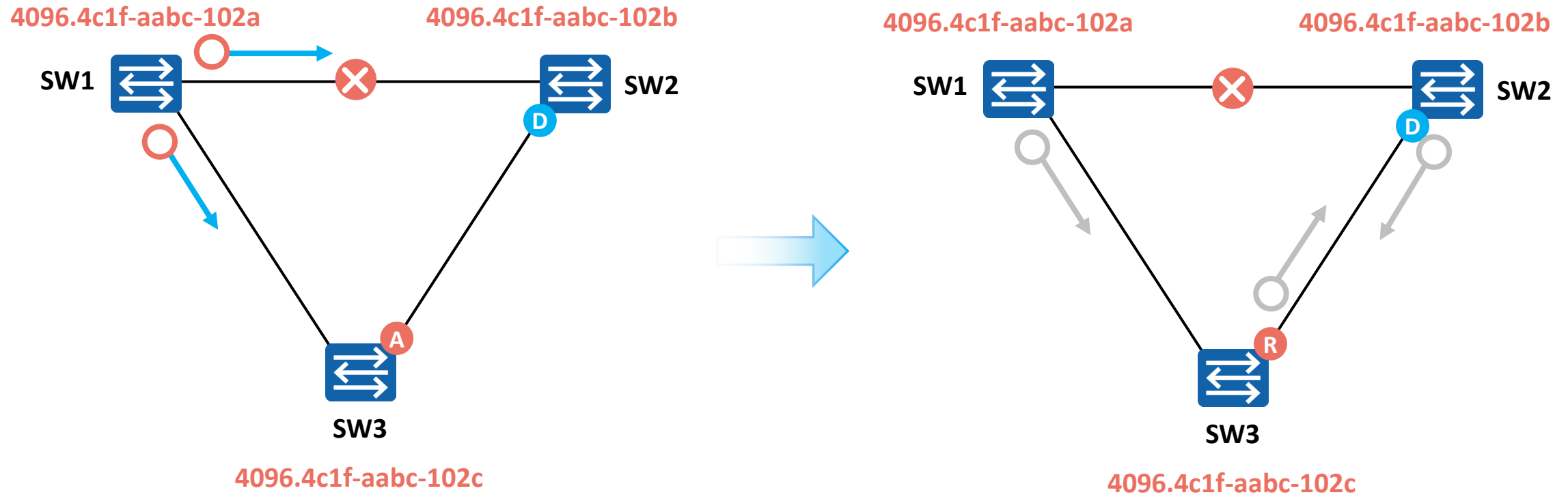
On a stable network, when SW2 detects that the link of the root port is faulty, the alternate port of SW2 enters the Forwarding state after twice the value of the Forward Delay timer (the default value is 15s).

- After SW2 detects a fault on the direct link, it switches the alternate port to the root port.
- If a direct link fails, the alternate port restores to the Forwarding state after 30s.



Topology Change: Indirect Link Fault

- When the indirect link fails, the alternate port on SW3 restores to the Forwarding state. It takes about 50s to recover from an indirect link failure.



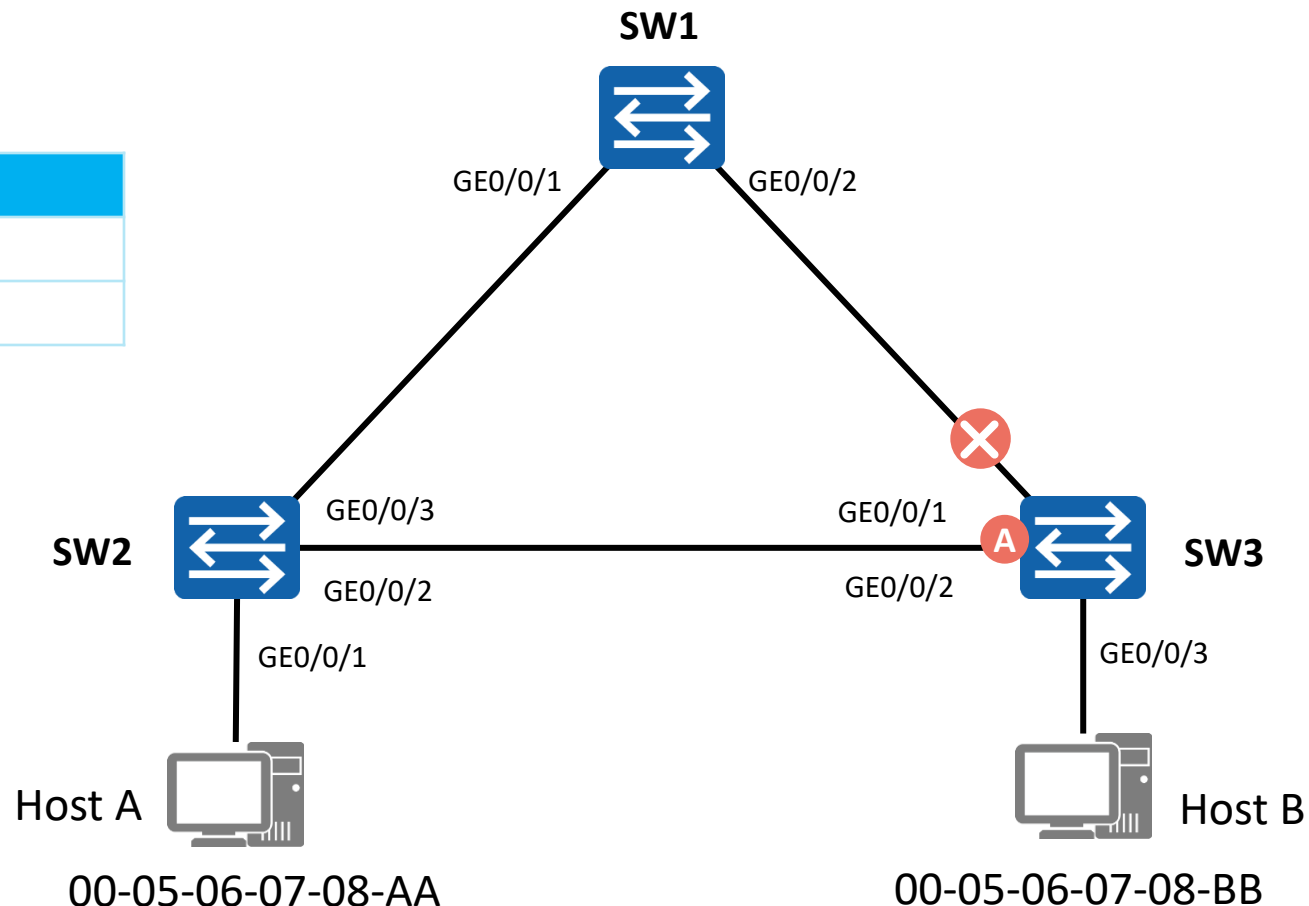


The MAC Address Table Is Incorrect Because the Topology Changes

MAC address table

MAC	Port
00-05-06-07-08-AA	GE0/0/1
00-05-06-07-08-BB	GE0/0/3

As shown in the figure, the root port of SW3 is faulty, causing the spanning tree topology to re-converge. After the spanning tree topology re-converges, Host B cannot receive frames sent by Host A. This is because switches forward data frames based on the MAC address table. By default, the aging time of MAC address entries is 300s. How is forwarding restored rapidly?



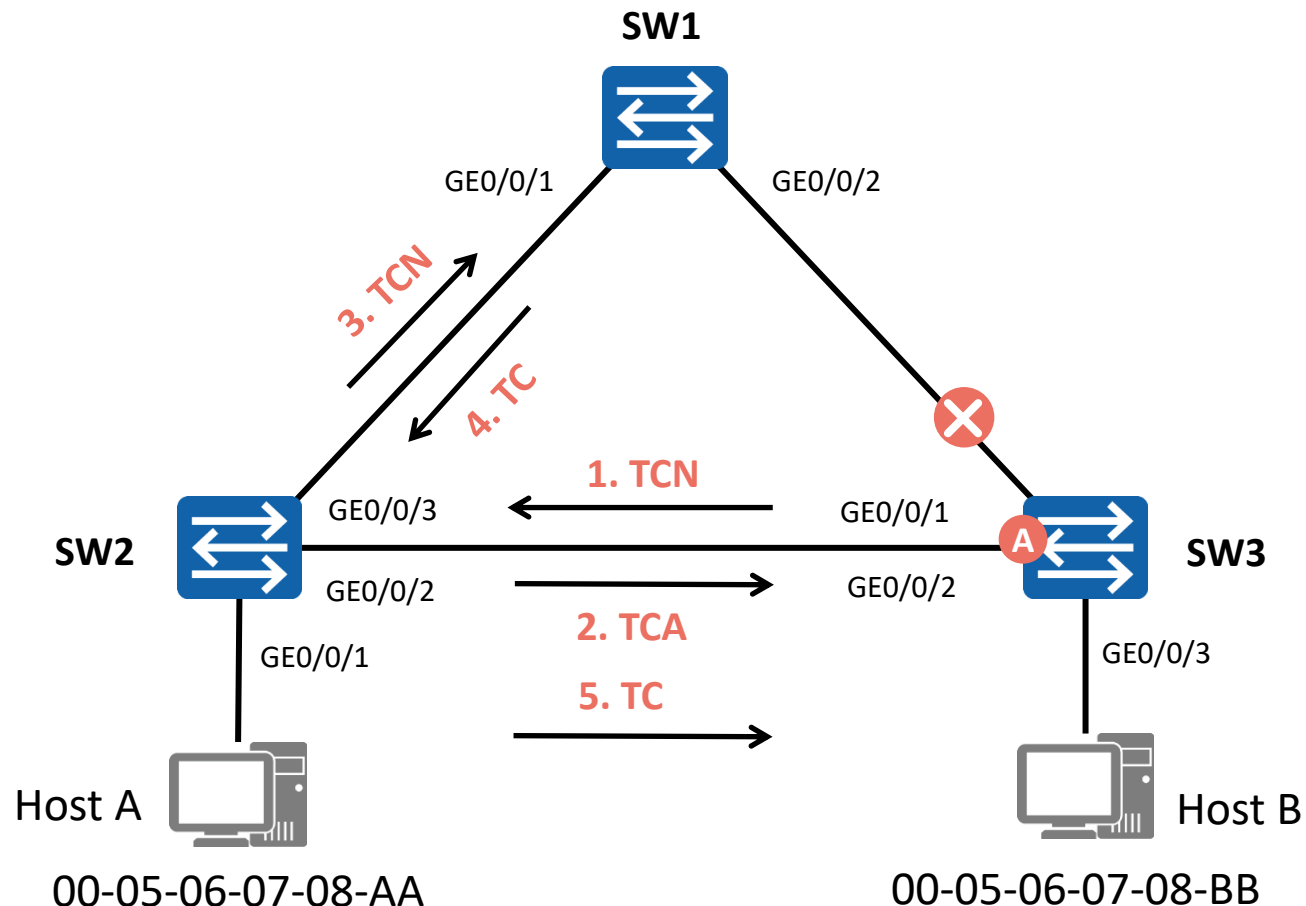


The MAC Address Table Is Incorrect Because the Topology Changes

MAC address table

MAC	Port
00-05-06-07-08-AA	GE0/0/3
00-05-06-07-08-BB	GE0/0/1
00-05-06-07-08-BB	GE0/0/2

- TCN BPDUs are generated when the network topology changes.
- Packet format: protocol identifier, version number, and type
- Topology change: The TCA and TC bits in the Flags field of configuration BPDUs are used.





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Basic STP Configuration Commands (1)

1. Configure a working mode.

```
[Huawei] stp mode { stp | rstp | mstp }
```

The switch supports three working modes: STP, RSTP, and Multiple Spanning Tree Protocol (MSTP). By default, a switch works in MSTP mode. On a ring network running only STP, the working mode of a switch is configured as STP; on a ring network running RSTP, the working mode of a switch is configured as RSTP.

2. (Optional) Configure the root bridge.

```
[Huawei] stp root primary
```

Configure the switch as the root bridge. By default, a switch does not function as the root bridge of any spanning tree. After you run this command, the priority value of the switch is set to 0 and cannot be changed.

3. (Optional) Configure the switch as the secondary root bridge.

```
[Huawei] stp root secondary
```

Configure the switch as the secondary root bridge. By default, a switch does not function as the secondary root bridge of any spanning tree. After you run this command, the priority value of the switch is set to 4096 and cannot be changed.



Basic STP Configuration Commands (2)

1. (Optional) Configure the STP priority of a switch.

```
[Huawei] stp priority priority
```

By default, the priority value of a switch is 32768.

2. (Optional) Configure a path cost for a port.

```
[Huawei] stp pathcost-standard { dot1d-1998 | dot1t | legacy }
```

Configure a path cost calculation method. By default, the IEEE 802.1t standard (**dot1t**) is used to calculate path costs. All switches on a network must use the same path cost calculation method.

```
[Huawei-GigabitEthernet0/0/1] stp cost cost
```

Set the path cost of the port.



Basic STP Configuration Commands (3)

1. (Optional) Configure a priority for a port.

```
[Huawei-GigabitEthernet0/0/1] stp priority priority
```

Configure a priority for a port. By default, the priority of a switch port is 128.

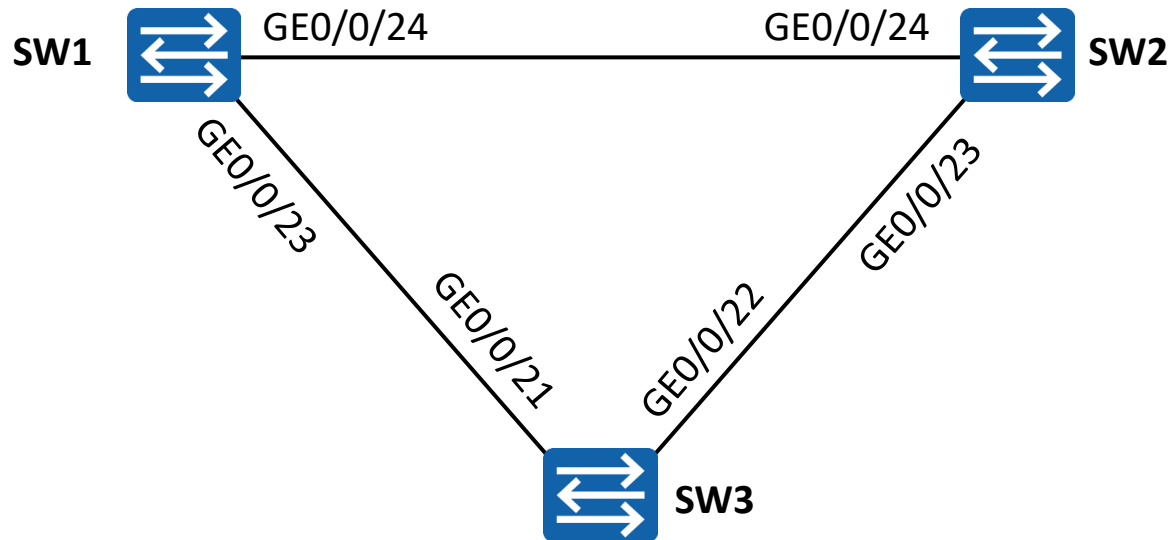
2. Enable STP, RSTP, or MSTP.

```
[Huawei] stp enable
```

Enable STP, RSTP, or MSTP on a switch. By default, STP, RSTP, or MSTP is enabled on a switch.



Case 1: Basic STP Configurations



- Deploy STP on the three switches to eliminate Layer 2 loops on the network.
- Configure SW1 as the root bridge and block GE0/0/22 on SW3.

SW1 configuration:

```
[SW1] stp mode stp  
[SW1] stp enable  
[SW1] stp priority 0
```

SW2 configuration:

```
[SW2] stp mode stp  
[SW2] stp enable  
[SW2] stp priority 4096
```

SW3 configuration:

```
[SW3] stp mode stp  
[SW3] stp enable
```



Case 1: Basic STP Configurations

Check brief information about STP states of ports on SW3.

```
<SW3> display stp brief
```

MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/21	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/22	ALTE	DISCARDING	NONE



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5. STP Advancement



Disadvantages of STP

- STP ensures a loop-free network but is slow to converge, leading to service quality deterioration. If the network topology changes frequently, connections on the STP network are frequently torn down, causing frequent service interruption.
- STP does not differentiate between port roles according to their states, making it difficult for less experienced administrators to learn about and deploy this protocol.
 - Ports in Listening, Learning, and Blocking states are the same for users because none of these ports forwards service traffic.
 - In terms of port use and configuration, the essential differences between ports lie in the port roles but not port states.
 - Both root and designated ports can be in Listening state or Forwarding state, so the port roles cannot be differentiated according to their states.
- The STP algorithm does not determine topology changes until the timer expires, delaying network convergence.
- The STP algorithm requires the root bridge to send configuration BPDUs after the network topology becomes stable, and other devices process and spread the configuration BPDUs through the entire network. This also delays convergence.



RSTP Overview

- RSTP defined in IEEE 802.1w is an enhancement to STP. RSTP optimizes STP in many aspects, provides faster convergence, and is compatible with STP.
- RSTP introduces new port roles. When the root port fails, the switch can enable the alternate port to obtain an alternate path from the designated bridge to the root bridge. RSTP defines three states for a port based on whether the port forwards user traffic and learns MAC addresses. In addition, RSTP introduces the edge port. The port connecting a switch to a terminal is configured as an edge port that enters the Forwarding state immediately after initialization, thus improving the working efficiency.



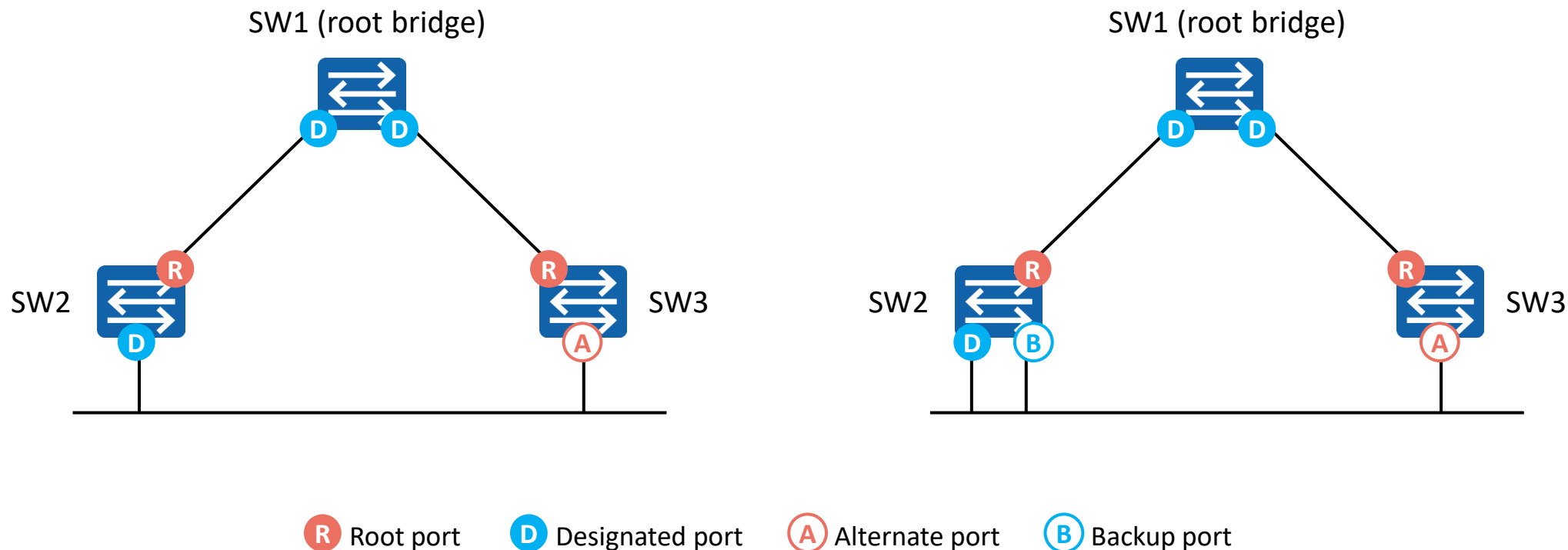
Improvements Made in RSTP

- RSTP processes configuration BPDUs differently from STP.
 - When the topology becomes stable, the mode of sending configuration BPDUs is optimized.
 - RSTP uses a shorter timeout interval of BPDUs.
 - RSTP optimizes the method of processing inferior BPDUs.
- RSTP changes the configuration BPDU format and uses the Flags field to describe port roles.
- RSTP topology change processing: Compared with STP, RSTP is optimized to accelerate the response to topology changes.



Port Roles in RSTP

- RSTP adds port roles to help understand RSTP and simplify RSTP deployment.

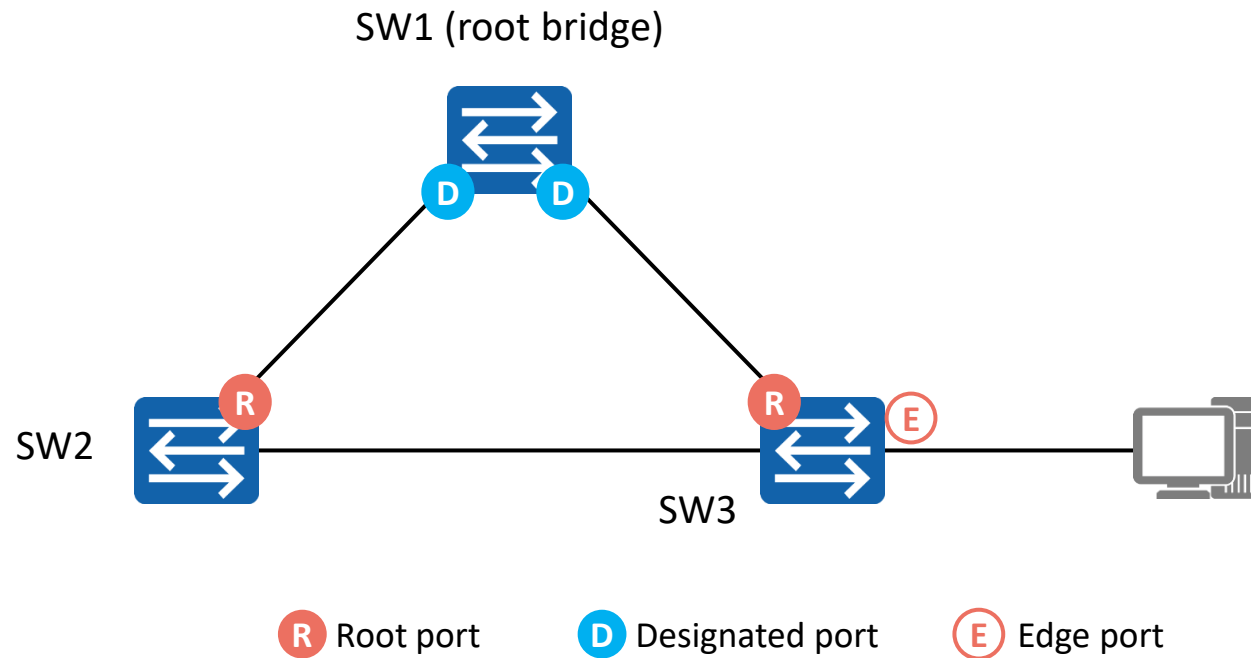


RSTP defines four port roles: root port, designated port, alternate port, and backup port.



Edge Port

- An edge port is located at the edge of a region and does not connect to any switching device.



Generally, an edge port is directly connected to a user terminal. The edge port can transition from the Disabled state to the Forwarding state.



Port States in RSTP

- RSTP deletes two port states defined in STP, reducing the number of port states to three.
 - If the port does not forward user traffic or learn MAC addresses, it is in Discarding state.
 - If the port does not forward user traffic but learns MAC addresses, it is in Learning state.
 - If the port forwards user traffic and learns MAC addresses, it is in Forwarding state.

STP Port State	RSTP Port State	Port Role
Forwarding	Forwarding	Root port or designated port
Learning	Learning	Root port or designated port
Listening	Discarding	Root port or designated port
Blocking	Discarding	Alternate port or backup port
Disabled	Discarding	Disabled port



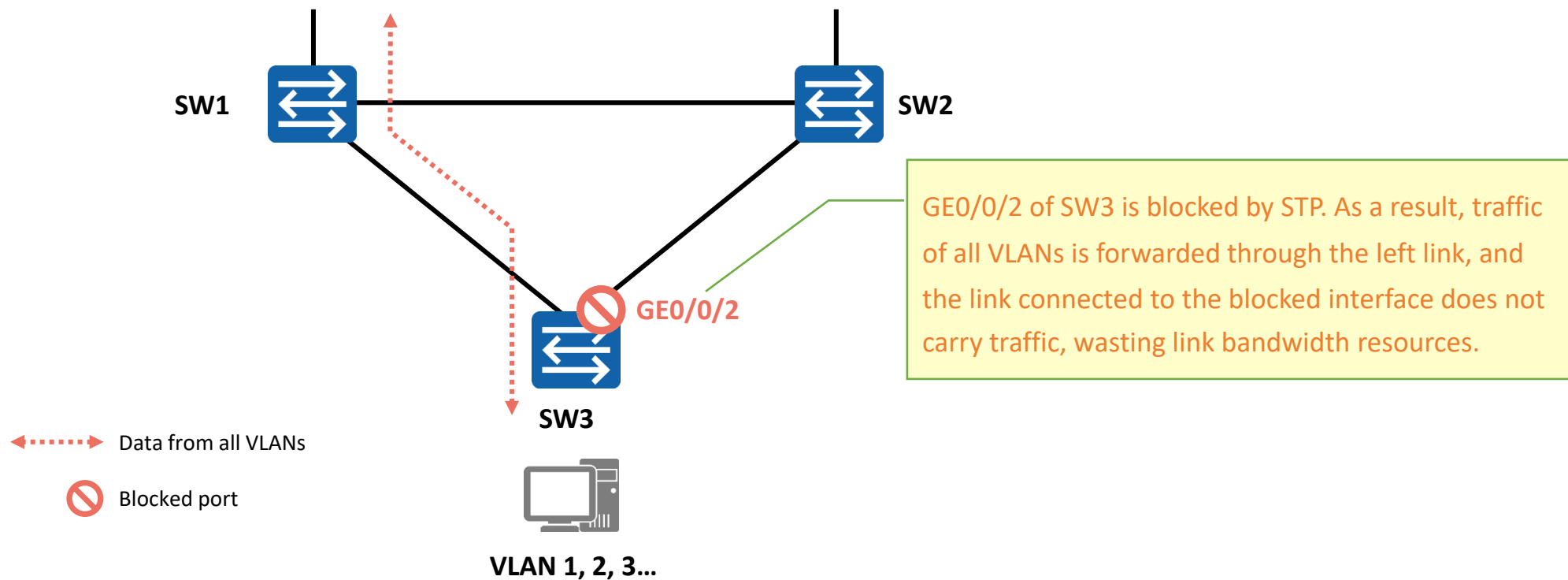
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Defects of STP/RSTP: All VLANs Share One Spanning Tree

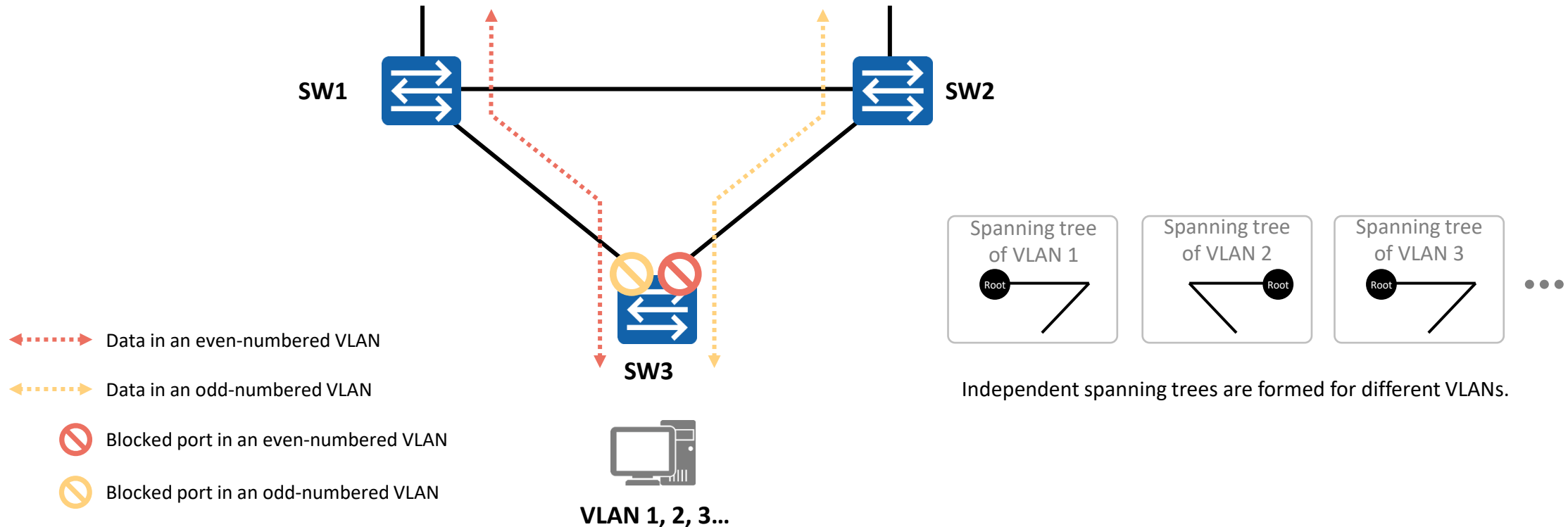
- RSTP, an enhancement to STP, allows for fast network topology convergence.
- STP and RSTP both have a defect: All VLANs on a LAN share one spanning tree. As a result, inter-VLAN load balancing cannot be performed, and blocked links cannot transmit any traffic, which may lead to VLAN packet transmission failures.





VBST

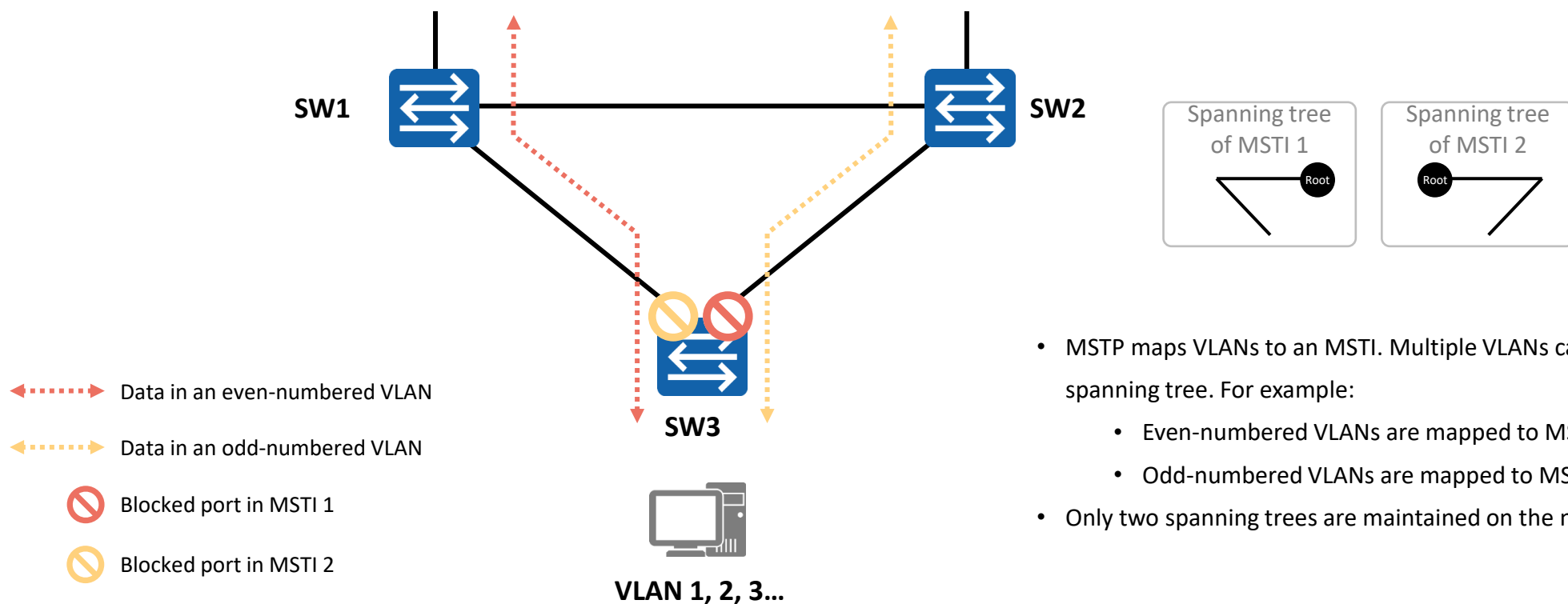
- Huawei provides the VLAN-based Spanning Tree (VBST). VBST constructs a spanning tree in each VLAN so that traffic from different VLANs is load balanced along different spanning trees.





MSTP

- To fix the defects, the IEEE released the 802.1s standard that defines the Multiple Spanning Tree Protocol (MSTP) in 2002.
- MSTP is compatible with STP and RSTP, and can rapidly converge traffic and provides multiple paths to load balance VLAN traffic.



- MSTP maps VLANs to an MSTI. Multiple VLANs can share one spanning tree. For example:
 - Even-numbered VLANs are mapped to MSTI 1.
 - Odd-numbered VLANs are mapped to MSTI 2.
- Only two spanning trees are maintained on the network.



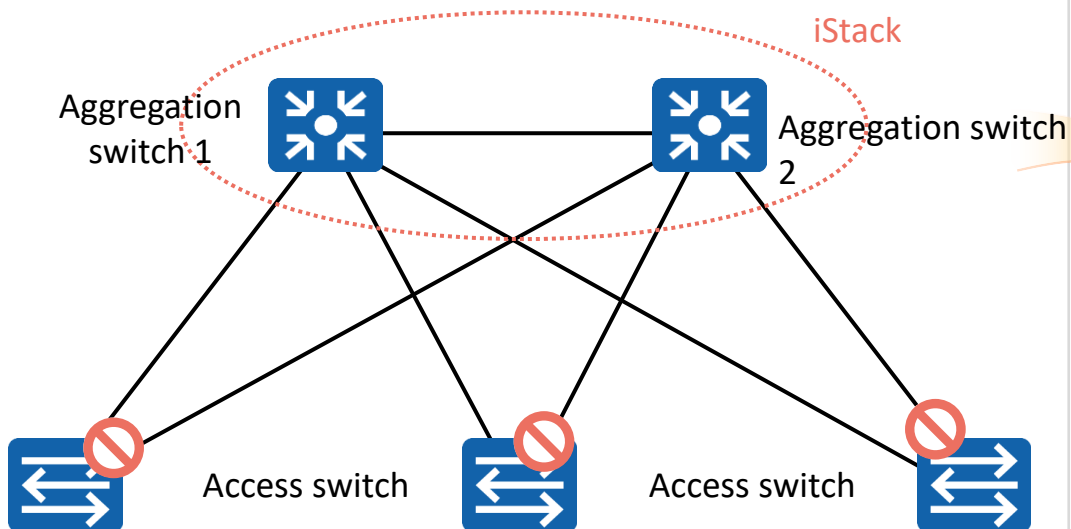
MSTP Overview

- MSTP divides a switching network into multiple regions, each of which has multiple spanning trees that are independent of each other.
- Each spanning tree is called a multiple spanning tree instance (MSTI).
- An MSTI is the spanning tree corresponding to a set of VLANs.
- Binding multiple VLANs to a single MSTI reduces communication costs and resource usage.
- The topology of each MSTI is calculated independently, and traffic can be balanced among MSTIs.
- Multiple VLANs with the same topology can be mapped to a single MSTI. The forwarding state of the VLANs for an interface is determined by the interface state in the MSTI.



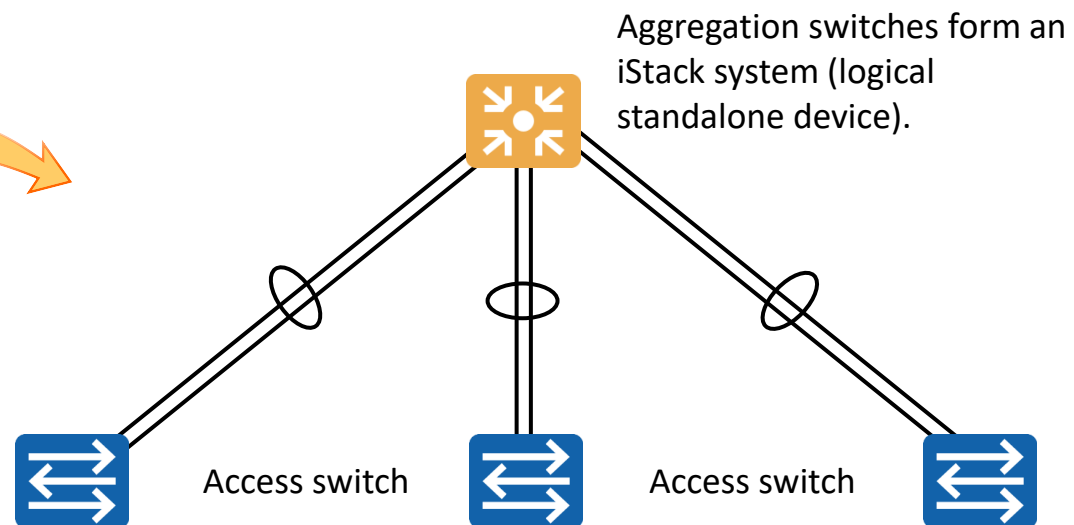
Stack and Tree Networking of Campus Networks

Traditional STP Networking



Two aggregation switches form a triangle Layer 2 loop with access switches, so STP must be deployed on the network. However, STP blocks ports on the network, causing a failure to fully utilize link bandwidth.

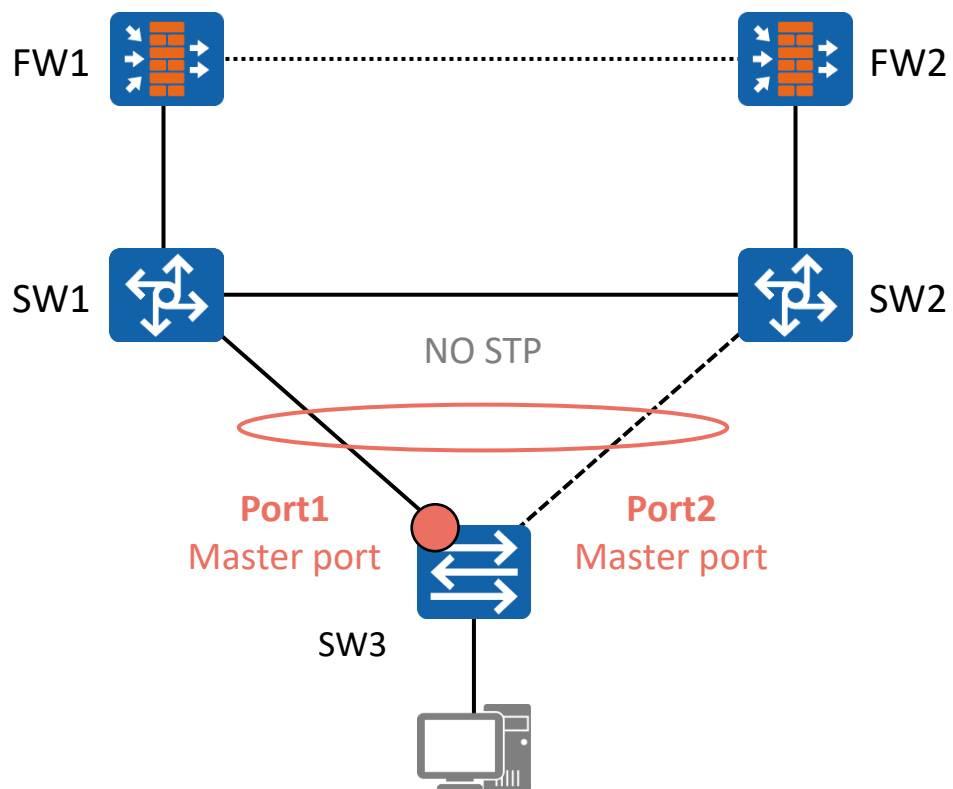
iStack Networking



Aggregation switches are stacked to form a single logical device, simplifying the network topology. In addition, link aggregation is deployed between aggregation switches and access switches to simplify the network topology to a tree topology, eliminating Layer 2 loops and improving link bandwidth utilization.



Smart Link



 Smart Link group

 Active status

Smart Link is tailored for dual-uplink networking.

- Smart Link is deployed on two switches where a host is dual-homed. When the network is normal, one of the two uplinks is active, and the other is in standby state (does not carry service traffic). In this way, a Layer 2 loop is eliminated.
- When the active link is faulty, traffic is switched to the standby link in milliseconds. This ensures proper data forwarding.
- Smart Link is easy to configure.
- Smart Link does not involve protocol packet exchange, therefore greatly improving speed and reliability.



Quiz

1. (Single Choice) Which statement about the STP port state is false? ()
 - A. The blocked port does not listen to or send BPDUs.
 - B. A port in Learning state learns MAC addresses but does not forward data.
 - C. A port in Listening state keeps listening to BPDUs.
 - D. If a blocked port does not receive BPDUs within a specified period, the port automatically switches to the Listening state.



Summary

- STP prevents loops on a LAN. Devices running STP exchange information with one another to discover loops on the network, and block certain ports to eliminate loops. With the growth in scale of LANs, STP has become an important protocol for a LAN.
- After STP is configured on an Ethernet switching network, the protocol calculates the network topology to implement the following functions:
 - Loop prevention: The spanning tree protocol blocks redundant links to prevent potential loops on the network.
 - Link redundancy: If an active link fails and a redundant link exists, the spanning tree protocol activates the redundant link to ensure network connectivity.
- STP cannot meet requirements of modern campus networks. However, understanding the working mechanism of STP helps you better understand the working mechanism and deployment of RSTP and MSTP.

The image features a blue-tinted background with silhouettes of several groups of business professionals. On the left, a group of four people is gathered around a table, looking at documents. In the center, two groups are visible: one of three people standing and talking, and another of four people, with one person holding a large document or tablet. On the right, a group of four people is engaged in conversation. The entire scene is set against a backdrop of modern skyscrapers, creating a professional and corporate atmosphere.

Thank You
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